

FireAtlas — Winning Plan and Accountability Tracker

Challenge: NASA Space Apps 2026 · *Harmonization of MODIS and VIIRS Hot Spots* **Document version:** 1.0 · 29 September 2026 ·
Baseline audit score **31.8 / 100** (implementation) · **≈65 / 118** (local-event rubric estimate) **Companion files (same folder):**
SCORECARD.md (the live tracker an LLM updates) · src/*.md (this document's source) · scripts/build_winning_plan.py (rebuilds this PDF)

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0. How to use this document

This document is written for two readers:

1. **You (the team lead).** Read sections 0–4 once. Then use SCORECARD.md every day.
2. **An executor LLM** (any coding model). Give it the **Executor System Prompt** in section 0.3, then one **Task Master Prompt** at a time from sections 4–8. Never give it the whole plan at once and say "do everything".

0.1 The order of work (never skip a gate)

Order	Block	Section	Why it comes here
1	Gate 0 — eligibility and timeline	1	Space Apps rules may forbid work started before 14 Nov 2026. Nothing else matters if the project is ineligible.
2	Gate 1 — repository integrity	5 (C12-T1...T3)	The best work is uncommitted. A judge who clones today sees an older product.
3	Cleanup — remove what hurts	3 (target shape) and 4 (R1-R17)	Off-topic features lower Relevance and Validity scores.
4	Challenge core (Tier 1)	6 (C2, C1, C10)	The judged challenge is harmonization + calendar. This is 70-80 % of the win.
5	Interface, packaging and safety wording	6 (C11, C12-T4...T9, C5/C8/C9 Tier 1)	Judges see a 30-second video or 7 slides and a public link.
6	FireAtlas Scout replay (Tier 2)	7 (C3, C4, C6, C7 and Tier 2 of C5, C8, C9, C10)	Only after Tier 1 acceptance tests pass. Clearly labelled extension.
7	Submission package	8	Project page, video, slides, citations, AI disclosure, licence.
8	Tracking and re-scoring	9 (tracker, Auditor prompt) and 10 (timelines)	Keeps every claim tied to evidence.

0.2 Status words (use exactly these words in SCORECARD.md)

- NOT STARTED · IN PROGRESS · BLOCKED (reason) · DONE (evidence: ...) · DROPPED (reason)

A task is **DONE** only when **every** acceptance check in its checklist passes **and** the evidence (command output, test name, screenshot path, commit hash) is written next to it.

0.3 Executor System Prompt (paste this first into any coding LLM)

```
You are the executor for the FireAtlas repository (NASA Space Apps 2026, challenge "Harmonization of MODIS and VIIRS Hot Spots"). Repository root: the folder that contains pyproject.toml and the fireatlas/ package. Python 3.10+, dependency manager: uv. Run tests with: uv run python -m unittest discover -s tests

NON-NEGOTIABLE RULES
1. Never invent data, numbers, dates, URLs, API fields, dataset versions or results.
   If a value is not in a file you read or a command you ran, write "UNKNOWN" and stop to ask.
2. Before editing a file, read it. Quote the exact lines you will change (path:line).
3. Do only the task you were given. Do not refactor unrelated code. Do not rename public API fields unless the task says so.
4. Every new behaviour needs a unit test in tests/. Run the full test suite before you say you are done. Paste the final "Ran N tests ... OK" line.
5. Never mix synthetic and authentic data. Synthetic outputs must carry demo=true / a "SYNTHETIC" label. Authentic outputs must carry source IDs and file hashes.
6. Never state that the app predicts, validates, authorizes, dispatches, or guarantees anything unless a test and a validation file prove it. Hotspots are not perimeters, burned area, or proof of no fire.
7. Never produce flight routes, waypoints, drone altitudes, ignition lines, evacuation zones or "safe" areas. Link to official sources instead.
8. Never commit secrets (FIRMS MAP_KEY, Earthdata tokens). Keys stay in environment variables or ~/.config/fireatlas/.
9. If a step says "VERIFY", you must confirm it by running a command or reading the official page. If you cannot, mark the task BLOCKED and explain what is missing.
10. When finished, output a REPORT block exactly in this format:
    TASK: <id>
    STATUS: DONE | BLOCKED (reason)
    FILES CHANGED: <path list>
    TESTS: <command> -> <result line>
    EVIDENCE: <screenshots / JSON outputs / commit hash>
    SCORECARD UPDATE: <which rows in docs/winning-plan/SCORECARD.md to change and to what>
    RISKS / FOLLOW-UPS: <bullets>
```

0.4 How each task in this plan is written

Every task has the same parts. If a part is missing, treat the task as not ready.

- **ID** (for example C2-T3) — used in SCORECARD.md and commit messages (C2-T3: add sensor availability ledger).
- **Goal** — one sentence.
- **Files** — exact paths to create or edit.
- **Steps** — numbered, in order.
- **Acceptance checks** — the checklist that defines DONE.
- **Master prompt** — paste-ready text for the executor LLM.

0.5 Ground-truth facts (the executor must not contradict these)

These were measured during the 29 September 2026 audit. If code changes them, the SCORECARD must record the new value and the command that produced it.

Fact	Value	How it was measured
Authentic FIRMS standard rows in bundle	30,823 (MODIS_SP 8,151; VIIRS_SNPP_SP 22,672)	<code>archive.import_bundle</code> into an empty database
Bundle area and years	bbox -122.2, 38.8, -120.0, 41.0; July 2022 – June 2026 (84 complete months to Dec 2025)	<code>fireatlas/archive.py:26-28</code>
MODIS product versions	6.03 (Jul–Dec 2022), 61.03 (2023+)	SQL on <code>observations.product_version</code>
FIRMS <code>type</code> values in bundle	0: 30,620 · 2: 164 · 3: 39 (all currently counted)	SQL on <code>raw_json</code>
Park Fire case, 25 Jul 2024	MODIS 603 pixels → 415 cells; VIIRS 0 rows (S-NPP outage 24–28 Jul)	<code>/api/harmonization</code>
July 2025 enclosing box	474 cell-days vs median 152 (2022–2024), mixed MODIS versions	<code>/api/calendar</code>
CAL FIRE association	25 eligible incidents: 7 nearby detection, 18 miss	<code>/api/validity</code>
Native mask processing (Grove 2025)	VIIRS 3/3 processed; MODIS 0/17 (16 fail "found 0 Latitude layer", 1 missing file)	<code>python3 -m fireatlas.masks --case grove-2025 --store /tmp/...</code>
Test suite	81 tests pass	<code>uv run python -m unittest discover -s tests</code>
Committed state	Authentic pipeline (<code>archive.py</code> , <code>validity.py</code> , <code>harmonization.py</code> , <code>masks.py</code> , method page, NASA sample ZIP) is uncommitted ; local branch 2 commits ahead of <code>origin/main</code>	<code>git status</code> , <code>git show HEAD:fireatlas/web.py</code>
Scout features	Terrain analysis, fuels, weather, spread, sector ranking, aerial ingest, assimilation: absent	<code>grep</code> over <code>fireatlas/</code>
Official dates	Challenge statements 28 Oct 2026; hackathon 14–15 Nov 2026; submission closes 15 Nov 23:59 local	Space Apps FAQ
Sensor continuity	S-NPP delivery ends 1 Nov 2026; Aqua MODIS science collection ends ~Aug 2026; Terra ~Jan 2027	NASA/NOAA notices, LAADS transition PDF

1. Gate 0 — Eligibility and timeline (do this before any code)

Problem. The official Space Apps FAQ says: "*Teams are not allowed to begin working on the challenges prior to the hackathon.*" At submission, teams confirm "*all of your work is original.*" This repository's first commit is 27 Sep 2026; the hackathon is 14–15 Nov 2026.

G0-T1 — Get a written ruling

- **Goal:** know, in writing, whether existing code may be reused.
- **Steps:**
 1. Find your Local Event page on spaceappschallenge.org and the Local Lead's contact.
 2. Send the message produced by the prompt below. Also post the question in the official Space Apps help channel if your event uses one.
 3. Save the reply as `docs/winning-plan/evidence/G0-ruling.txt` (do not commit personal emails; redact addresses).
- **Acceptance checks:**
 - ☐ A written reply exists and is saved.
 - ☐ `SCORECARD.md` → "Gate 0" row set to `Branch A` (reuse allowed) OR `Branch B` (rebuild at event).

Master prompt (for any LLM, to draft the message):

Draft a short, polite email (under 180 words) to my NASA Space Apps 2026 Local Lead.
Facts: our team explored NASA FIRMS MODIS/VIIRS data and built a prototype web app in late September 2026, before the challenge statements (28 Oct) and the hackathon (14-15 Nov). The FAQ says teams may not begin working on challenges before the hackathon.
Ask: (1) May we reuse this prior code, data-processing scripts, or only ideas/notes?
(2) If reuse is allowed, must we disclose it on the project page, and how?
(3) If not allowed, may we keep the public datasets we downloaded?
Do not claim anything else about the project. Neutral tone. No marketing language.

G0-T2 — Choose the branch

Ruling	Branch	What it means for this plan
Reuse allowed (with disclosure)	Branch A	Execute sections 3–8 on this repository. Disclose prior work on the project page (section 8).
Reuse not allowed	Branch B	Freeze this repository as a private research notebook. Until 14 Nov, only write design notes, data-download lists and this plan (ideas are not code). At the event, create a new repository and rebuild Tier 1 from sections 6–7 using the master prompts. Budget: 36 working hours for Tier 1 only; skip Tier 2.
No answer by 25 Oct	Branch B by default	Safer option.

G0-T3 — Re-read the official statement on 28 Oct

- ☐ Copy the official challenge statement text into `docs/winning-plan/evidence/challenge-statement-2026.md` (with URL and retrieval date).
- ☐ For each requirement sentence, add a row to `SCORECARD.md` → `Requirement trace` with the feature that satisfies it.
- ☐ If the statement adds or removes a requirement (for example "global AOI", "fire weather", "download"), update section 6 tasks before continuing.

2. The two scoring systems

You are graded twice: by the **audit implementation score** (my 100-point system, which measures what actually works) and by **Space Apps judges**. Track both in `SCORECARD.md`.

2.1 Audit implementation score (100 points)

Score each category 0–5, then `points = weight × score ÷ 5`.

Score	Meaning	Evidence required before you may claim it
0	No artifact	—
1	Idea or documentation only	A document describing it
2	Mockup, hardcoded output, isolated experiment	Screenshot or script
3	Functional component; integration, authentic data or verification missing	Working code + unit tests
4	End-to-end on real historical data with reproducible checks	Authentic-data run + reproducible command + tests + screenshot
5	Plus independent validation, clear limits, reliable demo	Validation report file with metrics vs a baseline + public demo link + limits text

Hard caps (the auditor LLM must apply these):

- Any category using only synthetic data: **max 2**.
- Any category whose code is not committed and pushed: **max 2**.
- Any model/ranking without a stated method and a test: **max 2**.
- Any forecast without evaluation against held-out observations: **max 3**.
- Any output that could be read as an operational instruction (flight path, ignition line, evacuation zone): **category = 0** until removed.

#	Category	Wt	Now	Target (Tier 1 only)	Target (Tier 1 + 2)
C1	Challenge fit and working calendar	12	3	5	5
C2	Authentic NASA data, provenance, harmonization	12	3	5	5
C3	Terrain, fuels, vegetation, dated weather	8	1	2	4
C4	Explainable ranking of observation sectors	14	0	0	4
C5	UAS feasibility framing and airspace/command boundaries	7	1	2	4
C6	Dated aerial-observation analysis and state update	8	0	0	4
C7	Spread scenarios, uncertainty, input freshness	10	1	1	4
C8	Containment boundaries and operational safety	7	2	3	4
C9	Responder workflow, resident info, official-alert separation	6	2	3	4
C10	Validation, replay, leakage control, baselines	8	2	4	5
C11	Interface clarity and judge-demo reliability	4	3	5	5
C12	Build, docs, data access, reproducibility	4	2	5	5
	Total points	100	31.8	54.2	88.0

Arithmetic. Tier 1 only: 12 + 12 + 3.2 + 0 + 2.8 + 0 + 2.0 + 4.2 + 3.6 + 6.4 + 4 + 4 = **54.2**. Tier 1 + 2: 12 + 12 + 6.4 + 11.2 + 5.6 + 6.4 + 8.0 + 5.6 + 4.8 + 8.0 + 4 + 4 = **88.0**.

Important: the audit score weights FireAtlas Scout heavily (C3–C9 = 60 points), but Space Apps judges weight **challenge relevance**. A Tier-1-only project scoring 54/100 here can still win the challenge; a project with half-built Scout features and a weak calendar cannot. Never trade Tier 1 quality for Tier 2 breadth.

2.2 Space Apps judging (what actually wins)

Official criteria (2025 wording; 2026 guide due 13 Nov): Impact, Creativity, Validity, Relevance, Presentation, each scored 1–5 by local judges. **Global awards** are chosen later by NASA subject-matter experts (Best Use of Science, Best Use of Data, Best Use of Technology, Galactic Impact, Best Mission Concept, Most Inspirational, Best Use of Storytelling, Global Connection, Art & Technology, Local Impact).

Local-event rubric supplied by the team (screenshots in the repo root, source unverified — confirm with your Local Lead):

#	Criterion	Points	What it measures	Estimate now	Target
1	Impact	1–20	Scale/significance of problem, reach	10	17
2	Creativity	1–20	Originality of idea and execution	11	16
3	Validity	1–20	Scientific soundness, feasibility, usability	11	18
4	Relevance	1–20	Alignment to challenge; NASA data at the core	11	19

#	Criterion	Points	What it measures	Estimate now	Target
5	Presentation	1–20	Clarity, structure, storytelling, reach	12	18
6	Teamwork	1–5	Evidence of collaboration	2	5
7	User experience	1–5	5 = anyone can use it	3	5
8	NASA data usage	1–5	NASA open data + other sources clearly shown	4	5
9	Challenge category identified	0/1	2026 category named	1	1
10	Repository access	0/1	Public working link	0	1
11	Project page complete	0/1	NASA project page submitted	0	1
—	Women participation bonus	+5 %	Team composition	—	—
	Total	118		≈65	≈106

2.3 What past winners had in common (use as a checklist)

Sources: Space Apps awards pages 2019–2025, NASA winner announcements, winning teams' repositories.

- ☐ **Named NASA data at the centre** (for example FIRMS MCD14ML / VNP14IMGML, not "satellite data").
- ☐ **A working public demo link** plus a public repository (c.a.w.s.t.o.n. 2019, Team I.O. GROW 2024).
- ☐ **One clearly defined user and one plain question** the tool answers (farmers, fire authorities, event planners).
- ☐ **Evidence of validity:** statistical test, expert interview or real user test (Starflock 2022 used 20 million FIREX-AQ rows and a statistical finding; GROW 2024 tested with a fire-affected farm owner).
- ☐ **Explanatory visuals** instead of raw maps.
- ☐ **Honest limits** (c.a.w.s.t.o.n. labelled their mockups as mockups).
- ☐ **Competitor awareness:** another 2026 team (IGNIS) already published a 26-year, 9-region harmonized calendar using an overlap-ratio method. FireAtlas must beat it on **validity** (sensor-availability accounting, held-out validation, uncertainty, evidence tracing) and **presentation**, not just match it.

2.4 The winning story (memorize this; every screen must support it)

"MODIS is ending and Suomi NPP stopped delivering on 1 November 2026. Twenty-five years of NASA fire records are split across sensors that see fire differently. FireAtlas harmonizes MODIS and VIIRS into one burning-activity calendar for any study area, shows which days a sensor was missing, puts every number next to its uncertainty, and lets anyone trace a result back to the original NASA pixel."

Primary user: a land manager or fire-season planner comparing this season to the historical record. **Secondary users:** researchers and educators (method + evidence); responders (historical review brief, Tier 2 replay). **The one question:** *"Is this period unusual for this place, and can I trust the comparison across sensors?"*

2.5 Things to highlight in every demo, slide and page

- The **multi-year calendar heatmap** (years × days) for one area (C1).
- Sensor timeline with outages and mission ends** (C2): Terra 2000–, Aqua 2002–Aug 2026, S-NPP 2012–Nov 2026, NOAA-20 2020–, NOAA-21 2024–.
- The calibration and its held-out error** (C2/C10): "VIIRS sees about N× more 1 km cell-days than MODIS here; predicted within X % on years we held out" (N and X come from your own run — never from this document).
- "No detection ≠ no fire"** and **"degraded day"** markers (C2).
- Trace a day to its original pixels** (existing evidence drawer and recount) (C10).
- Park Fire 2024 as the concrete story, and one contrasting area (crop burning or savanna) for global impact.

3. Target product shape (what the finished site looks like)

3.1 Information architecture (final)

Page	Path	Purpose	Status
Calendar (home)	/	Hero + area picker + multi-year calendar + day evidence	Rebuild from existing index.html
Method & validation	/method.html	How harmonization works, calibration results, sensor timeline, recount, limits	Extend existing
Sources & citations	/data.html	Every dataset, version, URL, licence, retrieval date, hashes; AI-use disclosure	Rewrite existing
Fire replay (beta)	/replay.html	Tier 2 Scout: Park Fire uncertainty replay	New, only if Tier 1 done

Top navigation (exact labels, left to right): Calendar · Method & validation · Fire replay (beta) (hidden until C4–C7 pass) · Sources · Remove the "Labs & sources" dropdown.

3.2 Visual rules (apply everywhere)

- Keep the existing design system in `fireatlas/static/design.css` (graphite surfaces, ember accent, DM Sans + Space Grotesk). Do not add a new CSS framework.
- **Sensor colours (fixed, colour-blind safe, use CSS variables in design.css):** `--modis: #E69F00` (orange), `--viirs-snp: #56B4E9` (sky blue), `--viirs-noaa20: #009E73` (green), `--viirs-noaa21: #CC79A7` (pink), `--harmonized: #F0E442` (yellow), `--degraded: repeating_hatch #7a8a93`, `--unknown: #3b4a52`.
- Every number on screen shows its **unit** ("cell-days", "detections", "MW") and its **scope** (area, date range, series, UTC or local solar day).
- Every chart has a one-line caption beginning with **"What this shows:"** and one line beginning with **"What this does not show:"**.
- Synthetic content is never visible in the default site. If any remains (tests only), it must have a red `SYNTHETIC` badge.
- Performance budget: first meaningful paint < 2.5 s on a 4G profile; calendar API < 800 ms for preset areas; no request to a third-party service is required for the calendar to render.
- Accessibility: all controls keyboard-operable; colour is never the only signal (use hatch/icons for degraded/unknown); WCAG AA contrast.

4. Cleanup — things to remove, demote or keep (section R)

Each row is one task. Execute in order. Run the full test suite after each row; fix or delete tests that only covered removed features.

ID	Item	Location	Action	Why
R1	Global disaster casebook (15 disasters with death tolls, flame textures)	fireatlas/static/documented-fires.json; #globe-documented, #globe-documented-toggle in index.html (lines ~67, ~99-100); casebook code in globe.js; route in web.py ASSETS	Delete	Not NASA data, not related to harmonization, emotionally loaded death counts invite judge skepticism.
R2	Training Lab (fictional Alder Creek crew exercise)	training.html/js/css, training-state.js, training-store.js, training-offline.js, training-sw.js, fireatlas/training.py, /api/training/* in web.py, tests/test_training.py, tests/test_web.py::test_training_routes_and_causal_snapshots	Move to branch archive/training-lab, then delete from main	Off-challenge and synthetic; dilutes Relevance and Validity.
R3	Installable PWA / offline shell	install.html, install.js, manifest.webmanifest, app-sw.js, offline.html, icons, tests/test_app_sw.py, docs/MOBILE_APP.md, scripts/verify_mobile_app.py	Delete (keep icons only if used as favicon)	Service-worker caching can serve stale data during judging; no rubric value.
R4	Synthetic "Explore demo" mode in the UI	Demo toggle and demo=1 links in index.html/app.js/study-ui.js/story.js; --judge-demo in web.py:531-534; scripts/launch_judge_demo.ps1	Remove from UI and launcher. Keep fireatlas/demo.py and presentation.py only for tests	Authentic data exist; synthetic mode is a validity liability.
R5	Guided tour built on synthetic July 2015	story.js, #story-dock	Replace with a 4-step tour on authentic data (C11-T5)	Current tour uses fabricated data.
R6	EONET "World snapshot" section	#live-events in index.html (~line 228), events.js, fireatlas/events.py, /api/events	Remove from home. Optionally list EONET on /data.html as a separate reported-events source	Not part of the harmonized record; live network dependency.
R7	7-day NRT global globe as the hero	globe.js, earth-embed.js, /api/globe*, NRT database rows	Demote: hero shows the sensor timeline and harmonized-archive coverage areas (C11-T1, C11-T3). Keep NRT import code for the "recent season" extension only	NRT data are rolling snapshots, not the historical record; 866,956 local rows are not reproducible from a clone.
R8	ArcGIS terrain globe	terrain-earth.html, earth-embed.js ArcGIS path	Remove from home. Reuse only on /replay.html if Tier 2 needs terrain visuals	Third-party network dependency; failed to load in audit; not analysis.
R9	14.5 MB earth.html and fireatlas_terrain_fixed.html at repo root	root; /earth.html route (web.py EARTH_MODEL); tests/test_web.py:44-51	Delete files, route and tests	Heavy, unused by the core story.
R10	Research Lab as a separate page	research.html/js/css, fireatlas/research.py	Merge the overlap table and candidate groups into /method.html → "Advanced analysis" collapsible; delete the standalone page and nav link	Fewer pages, one story.
R11	NOAA HMS in the calendar sensor switch	data-series="hms-viirs" button in index.html; SERIES["hms-viirs"] stays in core.py	Remove from the main switch. Use HMS only as an independent cross-check in validation (C10-T4)	It is NOAA, not the MODIS/VIIRS pair; confuses the story.

ID	Item	Location	Action	Why
R12	"Imported product" dropdown with NRT series	index.html sensor view row	Move to /data.html	Keeps the calendar focused on harmonized series.
R13	Stale presentation guide and screenshots (synthetic era)	docs/presentation/*	Delete after the new video/slides exist (section 8)	Screenshots show removed features.
R14	Root-level phone screenshots of the rubric	825258880_....jpg, 825260014_....jpg	Move to docs/winning-plan/evidence/local-rubric-*.jpg (or keep out of Git)	Clutter; keep as evidence of the rubric source.
R15	Project plan PDF at root	NASA_Space_Apps_Fire_Atlas_Project_Plan.pdf	Move to docs/archive/	Clutter.
R16	Hardcoded S-NPP gap	app.js:383-417,456; validity.py:34-38 source_notice	Replace by the data-driven availability ledger (C2-T3)	Only one outage is known to the app.
R17	Overlong README (281 lines, three datasets mixed) and contradictory PRD line 130	README.md, PRD.md	Rewrite (C12-T5)	Judges read the first screen only.

Master prompt for any R-row:

```
TASK R<n>. Read docs/winning-plan/src/03-cleanup.md row R<n>. Then:
1. List every file, route, test and HTML element that references the item
   (use ripgrep; paste the matches with path:line).
2. If the action is "Move to branch", create branch archive/<name> from main first
   and push it; then return to main.
3. Remove or change exactly those references. Do not touch unrelated code.
4. Delete tests that only exercised the removed feature; keep all other tests.
5. Run: uv run python -m unittest discover -s tests (must pass)
6. Start the server (uv run python -m fireatlas.web --port 8000), load /, /method.html,
   /data.html, and confirm there are no 404s or console errors
   (report the browser console output or curl status codes).
Output the REPORT block from the Executor System Prompt.
```

Checklist (all R rows):

- ☐ R1 · ☐ R2 · ☐ R3 · ☐ R4 · ☐ R5 · ☐ R6 · ☐ R7 · ☐ R8 · ☐ R9 · ☐ R10 · ☐ R11 · ☐ R12 · ☐ R13 · ☐ R14 · ☐ R15 · ☐ R16 · ☐ R17
- ☐ After all rows: rg -n "training|documented-fires|earth.html|install.html|demo=1" fireatlas/static returns only intentional matches (list them in the SCORECARD).

5. Gate 1 — Repository integrity (C12-T1 to C12-T3)

Do these before any feature work. They take under two hours.

C12-T1 — Commit and push the uncommitted work

- **Goal:** GitHub contains everything the local app uses.
- **Steps:**
 1. `git status --short` and save the output to the task report.
 2. Confirm no secrets: `rg -n "MAP_KEY=|api_key|token" --hidden -g '!*/.venv/**'` must show no real keys.
 3. Stage code, tests, docs and small samples: `git add fireatlas tests docs scripts pyproject.toml uv.lock README.md PRD.md .gitignore`.
 4. Check sample sizes: `du -h fireatlas/samples/*` — each file must be < 50 MB (GitHub hard limit is 100 MB).
 5. Commit: C12-T1: commit authentic MODIS/VIIRS pipeline, validity cases and method page. Push to origin/main.
- **Acceptance checks:**
 - ☐ `git status` shows no modified tracked files and no untracked files under `fireatlas/` or `tests/`.
 - ☐ `git log origin/main -1` equals local HEAD.
 - ☐ Repository is **public** on GitHub (Settings → General → Danger zone → visibility). Record the URL in SCORECARD.

C12-T2 — Fresh-clone smoke test and Linux launcher

- **Files:** `scripts/launch_demo.sh` (new), `scripts/smoke_test.sh` (new), `.github/workflows/ci.yml` (new).
- **Steps:**
 1. `scripts/launch_demo.sh`: `#!/usr/bin/env bash, set -euo pipefail, uv sync, then uv run python -m fireatlas.web --db data/fireatlas.sqlite3 --port "${PORT:-8000}"`.
 2. `scripts/smoke_test.sh`: clones the repo into a temp directory, runs `uv sync`, starts the server on port 8799 in the background, waits for `/api/meta` to return HTTP 200 (max 60 s), then checks: `/` returns 200; `/api/calendar?year=2024&series=joint&bbox=-122,39.5,-121.3,40.5` returns JSON whose `july_detected_cell_days` equals the value recorded in SCORECARD "Ground truth"; kills the server; exits non-zero on any failure.
 3. CI workflow on `ubuntu-latest`: checkout, install `uv` (`astral-sh/setup-uv`), `uv sync`, `uv run python -m unittest discover -s tests, node --check` on every `fireatlas/static/*.js`.
- **Acceptance checks:**
 - ☐ `bash scripts/smoke_test.sh` exits 0 on a machine that never had the repo.
 - ☐ CI badge green on GitHub; badge added to README.

C12-T3 — Declare every dependency

- **Steps:**
 1. Add to `pyproject.toml` an optional extra: `[project.optional-dependencies] masks = ["numpy>=1.24,<3"]` and document that GDAL (with HDF4 and netCDF drivers) must come from the system (`apt install gdal-bin python3-gdal` or `conda-forge`). Do not `pip-install GDAL` in CI.
 2. `fireatlas/masks.py` must fail with a clear message if `osgeo` is missing: "Native mask processing needs GDAL with HDF4 support; see docs/DATA.md".
 3. Pin upper bounds already present; run `uv lock` and commit `uv.lock`.
- **Acceptance checks:**
 - ☐ `uv sync` on a clean machine works without GDAL; core tests pass.
 - ☐ Mask tests are skipped (not failed) when GDAL is missing (`unittest.skipUnless`).

Master prompt (Gate 1):

TASK C12-T1..T3. Follow `docs/winning-plan/src/04-gate1-and-data.md` section 5 exactly. Do not change application behaviour. Paste `git status` before and after, the push result, the smoke-test output, and the CI run URL. Output the REPORT block.

6. Tier 1 — Challenge core

C2 — Authentic NASA data, provenance and scientifically defensible harmonization (weight 12, now 3, target 5)

Why it matters: Validity and Relevance are 40 of the 100 headline judge points. This category is where FireAtlas can be clearly better than competitors.

Definitions used everywhere (write them into docs/HARMONIZATION_METHOD.md):

- **Source series:** MODIS_SP (Terra + Aqua, Collection 6.1 standard, MCD14ML), VIIRS_SNPP_SP (VNP14IMGML, Collection 2 standard), VIIRS_NOAA20_SP (VJ114IMGML, standard). NRT series never enter baselines.
- **Cell-day:** a 1 km EPSG:6933 grid cell with at least one detection centroid of the given series on that day (existing definition, `core.py:175-212`).
- **Harmonized unit: "S-NPP-equivalent cell-days".** S-NPP is the middle of the record (2012–2026) and bridges MODIS (past) and NOAA-20 (future).
 - 2012-01-20 → 2026-10-31: observed VIIRS_SNPP_SP cell-days when S-NPP is available that day.
 - Before 2012-01-20: MODIS_SP cell-days × $r(m)$ where $r(m)$ is the month-of-year VIIRS/MODIS ratio (C2-T5).
 - S-NPP missing day in 2012–2019: MODIS × $r(m)$, flagged `fallback-modis`.
 - S-NPP missing day in 2020+ or after 2026-11-01: NOAA-20 cell-days × $q(m)$ where $q(m)$ is the S-NPP/NOAA-20 ratio from their overlap, flagged `fallback-noaa20 / successor-noaa20`.
 - Every harmonized value carries `source_used`, `estimate_type` (observed | scaled), and a 95 % interval (equal to the value when observed).

C2-T1 — Generalize the FIRMS archive importer (multi-region, multi-year)

- **Files:** `fireatlas/archive.py`, new `fireatlas/regions.py`, `tests/test_archive.py`.
- **Steps:**
 1. Create `fireatlas/regions.py` with a `REGIONS` dict. Each region: `id`, `name`, `bbox` (west, south, east, north), `fire_regime` (text), `why` (one sentence). Start with these five; the executor must **VERIFY each bbox on a map** (e.g., open the bbox in `geojson.io`) and adjust only if it clearly misses the intended area:
 - `norcal` — Northern California (Park Fire 2024): -122.2, 38.8, -120.0, 41.0 (existing bundle area).
 - `punjab-haryana` — crop-residue burning, India: 73.8, 29.5, 77.6, 32.6.
 - `rondonia` — Amazon deforestation fires, Brazil: -66.8, -13.7, -59.8, -7.9.
 - `top-end-au` — tropical savanna, Northern Territory, Australia: 129.0, -17.0, 138.0, -11.0.
 - `california` — whole state (optional, large): -124.5, 32.5, -114.1, 42.0.
 2. Replace the fixed constants `BBOX` and `EXPECTED_YEARS` (`archive.py:26-28`) with arguments `--region <id>` and a `request.json` file inside each downloaded `DL_FIRE_*` folder. `request.json` fields (typed by the human from the FIRMS request e-mail): `request_id`, `product` (MODIS_SP | VIIRS_SNPP_SP | VIIRS_NOAA20_SP), `start_date`, `end_date`, `area` (bbox or "country:XXX").
 3. A month is `complete_export = true` only if the whole month lies inside `[start_date, end_date]` **and** the region bbox lies inside the requested area. Otherwise import as partial.
 4. Keep all existing checks (NRT rejection, hashes, row counts). Remove the "file must start on July 1" rule (`archive.py:50-51`).
 5. Output bundle name: `fireatlas/samples/nasa_firms_<region>_<start>_<end>.zip` **only if < 50 MB**; otherwise write to `data/bundles/` (git-ignored) and publish as a GitHub Release asset (C12-T6).
- **Human data acquisition (cannot be done by the LLM):**
 1. Open `https://firms.modaps.eosdis.nasa.gov/download/` → "Create New Request".
 2. Area: draw or enter the region bbox (or choose the country for large regions).
 3. Date range and source, one request per product: MODIS Collection 6.1 **2000-11-01 → 2025-12-31**; VIIRS S-NPP 375 m **2012-01-20 → latest standard month**; VIIRS NOAA-20 375 m **2020-01-01 → latest standard month**. VERIFY on the form whether a date-range length limit exists; if so, split by year.
 4. Format CSV. Submit; download from the e-mail link; unzip into `NASA_data/<region>/DL_FIRE_*`; create `request.json` in each folder.
- **Acceptance checks:**
 - ☐ `uv run python -m fireatlas.archive build --region norcal NASA_data/norcal` reproduces the existing 30,823 rows for 2022-07→2026-06 when given the same inputs (regression test).
 - ☐ Test: a month partially outside the request dates is imported as partial.
 - ☐ Test: a region bbox outside the requested area raises `ValueError`.
 - ☐ SCORECARD "Data inventory" table lists, per region and product: years, rows, complete months, file SHA-256.

C2-T2 — Precomputed daily aggregates (fast, small, publishable)

- **Files:** new `fireatlas/aggregates.py`, `tests/test_aggregates.py`.
- **Steps:**
 1. For each region, series and UTC day compute: `cells_<series>` (count), `raw_<series>` (pixels), `frp_sum_<series>` (MW, sum of numeric `frp`), plus union/intersection counts for MODISuS-NPP and MODISnS-NPP.
 2. Compute each value in **8 variants**: day definition {`utc`, `solar`} × types {`veg` (type 0 or missing), `all`} × confidence {`all`, `no-low`}. `solar` day = UTC date of acquisition_UTC + longitude/15 hours (local solar time; no time-zone library needed). `no-low` removes VIIRS 1 and MODIS confidence < 30.
 3. Write `fireatlas/samples/aggregates/<region>.json.gz` with schema `fireatlas-daily-aggregates-v1` (fields listed in Appendix A.1). Target size < 10 MB per region.

4. Aggregates must be reproducible: include input bundle hashes and the method version.

- **Acceptance checks:**

- ☐ For `norcal` July 2024 UTC/all/all, aggregate union cells equal `/api/calendar` joint values day by day (test).
- ☐ Solar-day test: a detection at 2024-07-25T06:30Z at lon -121.5 goes to solar date 2024-07-24 (06:30 – 8.1 h).
- ☐ Type test: a row with `type=2` is excluded under `veg` and included under `all`.

C2-T3 — Sensor-availability ledger (replaces the hardcoded outage)

- **Files:** new `fireatlas/availability.py`, `fireatlas/samples/sensor_notices.json`, `fireatlas/samples/availability/<region>.json`, `tests/test_availability.py`; edit `core.py`, `harmonization.py`, `app.js` (delete `app.js:383-417,456` special case), `validity.py:34-38`.

- **Steps:**

1. **Overpass inventory from NASA CMR** (public, no login for metadata). For each region and product short name (MOD14 v061, MYD14 v061, VNP14IMG v002, VJ114IMG v002) query `https://cmr.earthdata.nasa.gov/search/granules.json` with `short_name`, `version`, `bounding_box=<region bbox>`, `temporal=<month start>,<month end>`, `page_size=2000`. Paginate using the CMR-Search-After response header (VERIFY in CMR API docs; `fireatlas/granules.py:27-47` already queries CMR and can be reused). Record each granule's `time_start`.
2. Count granules per UTC day and per solar day → `overpass_granules`. Store monthly JSON files so reruns resume.
3. `sensor_notices.json`: curated list of official notices. Each entry: `source_id`, `start_utc`, `end_utc` (or null), `type` (outage | partial-outage | mission-end | delivery-end), `url`, `retrieved_utc`, `quote` (≤ 25 words copied from the notice). Seed entries (all VERIFY by opening the URL and copying the quote):
 - S-NPP processing stop 2024-07-24 05:28Z → 2024-07-28 (FIRMS outages page; already cited in `validity.py`).
 - S-NPP NRT and standard outage, Apr 28 → May 2026 (Earthdata alert "data-outage-alert-suomi-npp-near-real-time-nrt-standard-products").
 - S-NPP data excluded 2026-08-03 02:00Z → 2026-08-06 16:24Z (LAADS alert 193182).
 - S-NPP delivery ends 2026-11-01 13:00Z (Earthdata alert "suomi-npp-data-product-delivery-cease-november-1-2026").
 - Aqua MODIS science collection ends ~2026-08 and Terra ~2027-01 (LAADS MODIS-VIIRS transition PDF).
4. Day state per source: `available` (≥ 1 granule, no notice) · `degraded` (partial-outage notice or granules < 50 % of that month's median daily count) · `missing` (0 granules or full outage notice) · `ended` (after mission/delivery end).
5. Calendar and harmonized series read this ledger. A joint/harmonized day with any required source not `available` gets `quality = "degraded"` and the UI hatches it.
6. Remove the hardcoded Park gap in `app.js` and `validity.py`; the Park 24-28 Jul flags must now come from the ledger.

- **Acceptance checks:**

- ☐ Park 2024-07-24..28 shows VIIRS_SNPP_SP degraded/missing from data, with the notice URL, and **no** literal 2024 date check remains in `app.js` (`rg -n "snppGap" fireatlas/static` returns nothing).
- ☐ Test with a synthetic ledger: a missing S-NPP day makes the harmonized value scaled with `source_used = MODIS_SP`.
- ☐ Every notice has a URL and retrieved date (test validates schema).
- ☐ Optional metric `cells_per_overpass` available in the API (value ÷ overpass granules), labelled "sampling-normalized, not cloud-corrected".

C2-T4 — Filters and version rules

- **Files:** `core.py` (`_counts`, `calendar`), `harmonization.py`, `tests`.

- **Steps:**

1. Add parameters `types="veg"|"all"` (default `veg`: `thermal_anomaly_flag` in ('0', NULL)) and `confidence="all"|"no-low"` (default `all`) to `_counts` and every endpoint that calls it. `thermal_anomaly_flag` already stores FIRMS type (`core.py:198`).
2. Baseline eligibility: a prior year is used only if each source's product versions belong to the same **collection** as the target year. Collection mapping (write as a table in code, VERIFY against FIRMS attribute docs): MODIS versions starting 6.1 or 61 → C6.1; starting 6.0/6.03/6 without 1 → C6; VIIRS 2/2.0 → C2. Mixed-collection years are **excluded from the baseline and listed** in `excluded_baseline_years` with the reason.
3. `briefing.py:42-45`: status must be `baseline-not-comparable` when versions are mixed or fewer than 10 comparable baseline years exist for anomaly language (Tier-1 rule: "unusual" language needs ≥ 10 years).

- **Acceptance checks:**

- ☐ July 2025 `norcal` with the current bundle: 2022 is excluded (6.03), the briefing says "baseline not comparable", not "322 above median".
- ☐ The 203 type ≠ 0 rows no longer count under the default filter (test using the real bundle).

C2-T5 — Overlap calibration with uncertainty (the scientific heart)

- **Files:** new `fireatlas/calibration.py`, `tests/test_calibration.py`, outputs `fireatlas/samples/calibration/<region>.json`.

- **Method (implement exactly; no other statistics libraries needed):**

1. **Overlap window** for MODIS+S-NPP: complete months from 2012-02 through the last complete month of both, excluding days where either source is not available in the ledger.
2. For each overlap year `y` and month-of-year `m`: $M[y,m] = \text{sum of MODIS daily cell-days on available days}$; $V[y,m] = \text{the same for S-NPP on the same days}$ (use variant `utc/veg/all`, plus repeat for others).
3. **Month-of-year ratio**: $r(m) = \sum_y V[y,m] / \sum_y M[y,m]$. Valid only if $\sum_y M[y,m] \geq 30$; otherwise use the pooled annual ratio $r_{all} = \sum V / \sum M$ and set `ratio_basis = "annual-fallback"`. If $\sum M < 30$ overall, harmonization for that region is unavailable.
4. **Uncertainty**: bootstrap by year. Resample overlap years with replacement 1,000 times (Python `random.Random(20261114)`); recompute $r(m)$; the 95 % interval is the 2.5th and 97.5th percentiles.
5. **Estimate for a MODIS-only month**: $\hat{V} = M \times r(m)$, interval $[M \times r_{lo}(m), M \times r_{hi}(m)]$. Daily estimates use the same monthly ratio.

6. **Leave-one-year-out (LOYO) validation:** for each overlap year, refit $r(m)$ without it, predict that year's $V[y,m]$ from its $M[y,m]$, and report: (a) median absolute log error $\text{median } |\ln((\hat{V}+1)/(V+1))|$ over months, (b) annual total absolute % error, (c) interval coverage (% of months where V falls inside the predicted interval).
7. **Baselines to beat** (report the same metrics): (i) *no harmonization* ($r = 1$), (ii) *single pooled ratio* (r_{all} , no month-of-year).
8. **Stability check:** Spearman rank correlation between year and annual ratio $\Sigma V_y / \Sigma M_y$, with a permutation p-value (5,000 permutations, same seed). If $p < 0.05$, set `ratio_trend_flag = true` and show a warning ("the MODIS/VIIIRS relationship changed over time; older estimates carry extra uncertainty").
9. Repeat steps 1–8 for **S-NPP ↔ NOAA-20** over 2020-01 → last common complete month to get $q(m)$.
10. Save `fireatlas-calibration-v1` JSON (Appendix A.2).

• **Acceptance checks:**

- ☐ Unit test with synthetic data where $V = 4 \times M$ exactly: $r(m) = 4$, interval width 0, LOYO error 0.
- ☐ Unit test: months with $\Sigma M < 30$ use `annual-fallback`.
- ☐ Real run for every region with overlap data; results table (region, r_{all} , LOYO median abs log error, coverage, beats-baselines yes/no) copied into SCORECARD.
- ☐ If the month-of-year model does **not** beat both baselines for a region, the UI uses the better baseline model for that region and says so. Never hide a failure.

C2-T6 — Provenance, citations and data licence text

• **Steps:**

1. Every new dataset (FIRMS archives per region, CMR inventories, notices, MCD64A1 in C10) gets a row in `docs/DATA.md` and on `/data.html`: name, product short name + version, provider, URL, date range, retrieval date, licence/terms, citation text, file SHA-256.
2. FIRMS acknowledgement text as published in the FIRMS FAQ (<https://www.earthdata.nasa.gov/data/tools/firms/faq>, read 29 Sep 2026): "*We acknowledge the use of imagery from the NASA LANCE FIRMS (<https://earthdata.nasa.gov/firms>), part of the NASA Earth Science Data and Information System (ESDIS).*" Re-check the FAQ before submission and copy any updated wording verbatim; image captions cite "NASA FIRMS".

• **Acceptance checks:**

- ☐ Test: every `source_id` present in the database has a row in `docs/DATA.md` (parse the markdown table).

Master prompt (use one task ID at a time):

TASK C2-T<n>. Read `docs/winning-plan/src/04-gate1-and-data.md`, section "C2-T<n>", and the files it lists. Implement exactly the listed steps and method. Where the text says VERIFY, verify by running a command or opening the official page, and quote what you found. Do not change the statistical method; if the data make the method impossible (for example too few overlap months), implement the documented fallback and report it. Add the listed unit tests. Run the full test suite. Then run the `real-data` command for region `norcal` and paste the key numbers. Output the REPORT block. Numbers you report must come from your command output.

C2 checklist: ☐ T1 importer · ☐ T2 aggregates · ☐ T3 availability ledger · ☐ T4 filters/versions · ☐ T5 calibration · ☐ T6 provenance · ☐ ≥ 3 regions with ≥ 20 years MODIS and ≥ 10 years S-NPP · ☐ LOYO beats both baselines in ≥ 3 regions (or failure shown honestly)

C1 — Direct challenge fit and a working burning-activity calendar (weight 12, now 3, target 5)

Goal state: for any preset region (and custom boxes inside it when running locally), a user sees 20+ years of harmonized activity as a calendar, can tell which periods were unusual and which days had missing sensors, and can open the pixels behind any day.

C1-T1 — Calendar API v2

- **Files:** new `fireatlas/calendar_v2.py`; route `/api/v2/calendar` in `web.py`; `tests/test_calendar_v2.py`.
- **Request parameters (validate each; reject unknown values with HTTP 400):**
 - `region` (id from `regions.py`) **or** `bbox` (must lie inside a loaded region; otherwise 400 "Area outside loaded historical record").
 - `series`: `harmonized` (default) | `modis` | `viirs-snpp` | `viirs-noaa20`.
 - `metric`: `cell_days` (default) | `detections` | `frp_sum`. `harmonized` supports `cell_days` only (the calibration is fitted on cell-days).
 - `day`: `utc` (default) | `solar`. `types`: `veg` (default) | `all`. `confidence`: `all` (default) | `no-low`.
 - `baseline`: YYYY-YYYY (default 2003-2022), minimum 10 years.
 - `years`: YYYY-YYYY range to return (default: all loaded).
- **Response (`fireatlas-calendar-v2`; full field list in Appendix A.3):** `meta` (series, metric, day, filters, region, bbox, method versions, calibration id, data class authentic); `years[]` each with `days[] = {date, value, low, high, estimate_type, source_used, quality, availability: {source: state}}; monthly[]` with `value, low, high, climatology: {p10, p50, p90, n_years}, percentile_rank, rank, flag; season per year (C1-T4); provenance[]; limits[]`.
- **Rules:**
 1. Days with no complete export → `value: null`, `quality: "unknown"`.
 2. Zero on a complete, available day → `value: 0`, `quality: "observed-zero"`, and the UI text "no detections recorded — not proof of no fire".

3. Percentiles use only baseline years that pass version and availability rules (C2-T4, C2-T3); list the years used.
4. flag values: unusually-high (≥ 90 th percentile), unusually-low (≤ 10 th), typical, insufficient-history (< 10 baseline years), not-comparable.

- **Acceptance checks:**

- ☐ Contract test: response validates against Appendix A.3 for `norcal`, every series and variant.
- ☐ Test: bbox outside loaded regions → 400 with the exact message.
- ☐ Performance: preset-region response < 800 ms on the dev laptop (log the timing in SCORECARD).

C1-T2 — Multi-year calendar heatmap (the signature visual)

- **Files:** `fireatlas/static/index.html` (replace the current monthly/day panels inside `#calendar-section`), new `fireatlas/static/calendar-heatmap.js`, `fireatlas/static/calendar.css`.
- **Layout (desktop ≥ 1100 px):**
 - Rows = years (newest at top), columns = day of year (1–366). Each cell 3 px wide $\times$ 14 px tall with 1 px gap; year label (Space Grotesk 12 px) at left; month ticks (Jan...Dec) on top.
 - Right of each row: a 120 px horizontal bar showing the **season window** (C1-T4) against the climatological window (thin outline).
 - Colour: sequential log scale from `--unknown-adjacent` dark to ember `#FFB163` (reuse `calendarHeat` in `app.js:374-379`); legend with 5 ticks and the unit.
 - `quality = degraded` → diagonal hatch overlay; `unknown` → flat `--unknown`; `estimate_type = scaled` → small dot in the cell's corner (legend: "estimated from another sensor").
 - Hover tooltip (and keyboard focus): 25 Jul 2024 (UTC) · 415 S-NPP-equivalent cell-days (estimated from MODIS; 95% 350–480) · S-NPP: missing (NASA notice) · MODIS: available. Numbers come from the API.
 - Click / Enter → opens the **day drawer** (C1-T6).
- **Mobile (< 760 px):** show one year at a time as a 7-column month-grid calendar (reuse the existing `#day-grid` rendering), with a year picker; same colours and states.
- **Render with `<canvas>`** for the grid ($\approx 9,000$ cells) and an invisible, focusable roving `<button>` for keyboard/screen readers that announces the tooltip text.
- **Acceptance checks:**
 - ☐ Screenshot `evidence/C1-T2-heatmap-norcal.png` shows ≥ 20 years; Park 2024 late July visibly hatched as degraded.
 - ☐ Keyboard: arrows move day/year, Enter opens drawer (manual test noted in SCORECARD).
 - ☐ Renders in < 300 ms after data arrive (performance.now() log).

C1-T3 — Monthly climatology and anomaly panel

- **Files:** `index.html` (panel under the heatmap), `calendar-heatmap.js` or new `climatology.js`.
- **UI:**
 - 12 monthly bars for the selected year, over a shaded p10–p90 band and a median line from the baseline years.
 - Bars coloured by flag: unusually-high ember with ▲ icon, unusually-low blue with ▼, typical neutral, insufficient-history hatched gray.
 - Under each flagged month a badge: Aug 2024 · 97th percentile · rank 1 of 21 years (2003–2022 baseline).
 - Caption lines "What this shows: monthly S-NPP-equivalent cell-days compared with the same month in the baseline years." / "What this does not show: burned area, fire counts, or risk."
- **Acceptance checks:**
 - ☐ Badge text is generated only from API fields (test: DOM text equals API values).
 - ☐ With < 10 baseline years the panel shows "Not enough comparable years for an 'unusual' label".

C1-T4 — Critical periods and season metrics

- **Files:** `fireatlas/calendar_v2.py` (season computation), UI in the heatmap right column + a "Season summary" card.
- **Method (write into `docs/HARMONIZATION_METHOD.md`):**
 1. For each year, daily harmonized values `x_d`. Cumulative share $C_d = \sum_{\{d \leq d\}} x / \sum_{year} x$ (skip null days; if $> 10\%$ of days are null, season metrics are unavailable).
 2. `season_start` = first day with $C_d \geq 0.10$; `season_peak` = centre of the 15-day window with the largest sum; `season_end` = first day with $C_d \geq 0.90$; `season_length` = `end` – `start` + 1.
 3. Climatological season = median start/peak/end over baseline years; report the shift of the selected year in days ("started 12 days earlier than usual").
 4. **Critical period** = runs of ≥ 5 consecutive days where the 15-day centred moving sum exceeds that day-of-year's baseline 90th percentile. List them with start/end dates and peak value.
 5. Regions spanning the southern-hemisphere season: compute the season on a **fire year** starting at the month of lowest climatological activity (store `fire_year_start_month` per region).
- **Acceptance checks:**
 - ☐ Unit test on a synthetic series with a known triangle-shaped season returns the exact start/peak/end.
 - ☐ Southern region test uses the rotated fire year.
 - ☐ UI lists critical periods with dates; clicking one scrolls the heatmap to it.

C1-T5 — Area selection

- **Files:** `index.html`, `app.js` (AOI controls), `fireatlas/regions.py`.
- **UI:**

- Region chips above the calendar (one per preset; label + fire regime, e.g., "Punjab-Haryana · crop-residue burning"). The selected chip is highlighted.
- "Draw a custom box" button (local server only): two clicks on the Leaflet map set the corners; the box must lie inside a loaded region; otherwise show "Outside the loaded historical record. Loaded areas: ...".
- Keep the text field `west,south,east,north` for power users; validate live.
- The static public site (C12-T6) shows presets only, with the note "Custom areas need the local install (see README)".
- **Acceptance checks:**
 - ☐ Every chip loads a calendar in < 2 s on the public site.
 - ☐ Custom box inside `norcal` works locally; outside shows the message.

C1-T6 — Day drawer (evidence for one day)

- **Files:** reuse `#evidence-section` logic in `app.js`; update copy.
- **Content, top to bottom:**
 1. Date + day definition (UTC or local solar) + region.
 2. Harmonized value with interval and `source_used`.
 3. Per-source rows: availability state (with notice link), raw pixels, cell-days, product versions.
 4. Small Leaflet map with the day's detections coloured by sensor (MODIS orange squares sized 1 km, VIIRS blue squares 375 m — drawn as approximate nadir squares with the note "approximate nadir size; real footprints grow toward swath edges").
 5. Table of up to 200 original records (existing `/api/observations`), each with a "raw row" expander showing the untouched CSV fields.
 6. Buttons: "Download this day (CSV)", "Recount this day" (calls the verifier, C10-T6).
- **Acceptance checks:**
 - ☐ For Park 2024-07-25 the drawer shows MODIS 603 pixels / 415 cells and S-NPP "missing (NASA notice)".

C1-T7 — Auto-generated insight text (no free-form LLM text)

- **Files:** `app.js` → replace `#insight-card` generator.
- **Templates (fill only from API fields; if a field is null, drop the sentence):**
 - `"{month} {year} in {region}: {value} S-NPP-equivalent cell-days ({percentile}th percentile of {n} baseline years, {baseline})."`
 - `"{k} day(s) this month had a missing or degraded sensor; their values are estimates."`
 - `"Season started {abs(shift)} days {earlier|later} than the {baseline} median."`
 - Always end with: `"Detections show where satellites recorded heat; they are not burned area or proof that no fire occurred elsewhere."`
- **Acceptance checks:** ☐ Unit test (Node) renders templates from a fixture and drops sentences with null fields.

C1-T8 — Exports

- CSV and JSON download of the current calendar view with a header block (`# series=...`, `region=...`, `method=...`, `calibration_id=...`, `generated_utc=...`, `citation=...`). Keep the study ZIP (`/api/study`) and add `calibration.json` + `availability.json` to it; extend `fireatlas/study.py` verifier to check them.
- **Acceptance checks:** ☐ `uv run python -m fireatlas.study <zip>` verifies a v3 bundle including calibration and availability files.

Master prompt (C1):

```
TASK C1-T<n>. Read docs/winning-plan/src/05-calendar-validation.md "C1-T<n>" and
Appendix A.3 (API schema) in docs/winning-plan/src/09-appendix.md. Read the existing
files listed before editing. Build exactly the described API/UI. All on-screen numbers
must come from API responses; never hardcode dates, counts or region-specific cases.
Keep the existing design system (fireatlas/static/design.css). Add tests (Python for API,
Node for pure JS functions). Run the full test suite. Start the server and capture a
screenshot of the new UI at 1440x900 and 390x844 into docs/winning-plan/evidence/.
Output the REPORT block.
```

C1 checklist: ☐ T1 API v2 · ☐ T2 heatmap · ☐ T3 climatology · ☐ T4 seasons/critical periods · ☐ T5 area selection · ☐ T6 day drawer · ☐ T7 insight text · ☐ T8 exports · ☐ a first-time user answers "Was August 2024 unusual here?" in < 60 s (user test, C11-T8)

C10 — Independent validation, retrospective replay, leakage control and baselines (weight 8, now 2, target 4 Tier 1 / 5 Tier 2)

C10-T1 — Publish the calibration validation

- Show the LOYO results (C2-T5) on `/method.html` → section "Does harmonization work?": a table per region (median abs log error, annual % error, interval coverage) for **month-of-year ratio vs single ratio vs no harmonization**, and a scatter plot (log-log) of predicted vs observed monthly S-NPP cell-days for held-out years with the 1:1 line.
- ☐ Numbers on the page equal `samples/calibration/<region>.json` (test).

C10-T2 — Sensor-transition step test

- **Question:** does the harmonized series jump at 2012 when S-NPP enters?
- **Method:** for each region, compare the observed annual ratio $\Sigma V_y / \Sigma M_y$ for 2012–2014 against the bootstrap 95 % interval of r_{all} fitted on 2015–2025 only. Report `step_ok = inside interval`. Also plot MODIS-only, naive stitched series (MODIS before 2012, S-NPP after), and harmonized series together for 2005–2020; the naive series will show the artificial jump.
- ☐ Figure `evidence/C10-T2-<region>.png` and table in SCORECARD.

C10-T3 — Independent reference: NASA MCD64A1 burned area

- **Data (human + script):** MODIS/Terra+Aqua Burned Area Monthly L3 Global 500 m, **MCD64A1 Collection 6.1**, from LP DAAC (Earthdata Login required). VERIFY the product page and version before download. Tools: NASA AppEEARS area request (region polygon, 2003–2025, band `Burn Date`) or direct HDF tiles.
- **Method:** monthly burned area km² inside each region (count pixels with `Burn Date > 0` × pixel area). Spearman correlation across all months 2003–2025 between burned area and (a) harmonized cell-days, (b) naive stitched series, (c) MODIS-only. Also correlation of **year-over-year changes** across the 2011→2012 boundary.
- **Honest interpretation text:** "Burned area is an independent product with different sensitivity (small fires and crop burns are often missed). Agreement supports, but does not prove, the harmonized record."
- ☐ Results table in SCORECARD and on `/method.html`. ☐ Script `fireatlas/reference_mcd64.py` + test on a tiny synthetic raster.

C10-T4 — Cross-check with NOAA HMS (existing data)

- Use the bundled NOAA HMS July 2021–2024 slice: daily Spearman correlation between HMS VIIRS cell-days and FIRMS S-NPP cell-days in the overlapping box. Present as "independent processing chain check".
- ☐ Table on `/method.html` "Advanced analysis".

C10-T5 — Fix native MODIS mask reading and finish the reconciliation

- **Observed failure:** 16/17 MODIS granules fail with "Expected one ('Latitude','latitude') layer, found 0" (`masks.py:97-99`).
- **Steps:**
 1. Run `gdalinfo NASA_data/fire_masks/MOD03.A2025185.0510.061.*.hdf | grep -i subdataset` and paste the output. Check `gdalinfo --formats | grep -i hdf4` to confirm the HDF4 driver exists.
 2. Adapt `native_layer` matching to the real subdataset names printed (do not guess). If HDF4 is missing, document the install and mark BLOCKED.
 3. Reprocess Park and Grove into `data/validity_masks.sqlite3`; reach the ≥ 98 % reconciliation target or report the actual rate.
 4. Human review: two team members independently review the 30 deterministic samples (existing queue) and record decisions in `docs/winning-plan/evidence/C10-T5-review.csv` (`sample_id`, `reviewer`, `decision`, `note`). Report agreement.
- ☐ Validation gates on `/method.html` show real statuses, not "pending".

C10-T6 — Recount verifier for calendar v2

- Extend `fireatlas/validation_check.py` to recompute, from exported aggregates + calibration + availability files, every harmonized daily value and interval (independent implementation of the formula in C2-T5 step 5). Expose a "Recount" button in the day drawer.
- ☐ Test: tampering with one aggregate value makes the recount fail with a clear message.

C10-T7 — Leakage rules (needed for C1 anomalies and Tier 2)

- Baselines only use years **before** the target year (already enforced for monthly baselines; enforce for percentiles and season climatology too).
- Calibration used for a pre-2012 estimate may use the full overlap period (it is a fixed method parameter, disclosed as such). Document this.
- ☐ Test: requesting 2015 with baseline `2003-2022` returns 400 or automatically restricts to `2003-2014` (choose one and document it; recommended: restrict and state it).

Master prompt (C10):

```
TASK C10-T<n>. Read docs/winning-plan/src/05-calendar-validation.md "C10-T<n>".
Implement the method exactly. Report metrics only from your own runs, with the command
used. If a metric is bad, report it and show it in the UI honestly; do not tune the
method on the held-out years. Add tests. Output the REPORT block.
```

C10 checklist (Tier 1): ☐ T1 · ☐ T2 · ☐ T3 · ☐ T4 · ☐ T5 · ☐ T6 · ☐ T7

C11 — Interface clarity and reliability of the judge demonstration (weight 4, now 3, target 5)

C11-T1 — Home page structure (final order, top to bottom)

1. **Header:** logo "fireatlas" (keep), nav from section 3.1, right side: GitHub link icon.

2. **Hero** (max 80 vh):
- Eyebrow: NASA SPACE APPS 2026 · HARMONIZATION OF MODIS AND VIIRS HOT SPOTS (names the challenge category — rubric item 9).
 - H1: 25 years of NASA fire detections. One calendar. (replace the year count with the real span once data are loaded).
 - Sub (≤ 30 words): MODIS and VIIRS see fire differently, and both are retiring. FireAtlas joins them into one burning-activity calendar—with missing days, uncertainty, and the original pixels in view.
 - Primary button Open the calendar → scrolls to calendar; secondary link How harmonization works → /method.html.
 - Right side: **sensor timeline graphic** (C11-T3) instead of the heavy globe. The globe may stay only if it loads in < 2 s and shows the preset regions (clicking a region selects it); otherwise remove it (R7/R8).
3. **Region chips + calendar heatmap** (C1-T2, C1-T5).
4. **Selected-year panel:** climatology bars (C1-T3) + season summary + critical periods (C1-T4) + insight text (C1-T7).
5. **Day drawer** (C1-T6), opens inline.
6. **"Can I trust this?" strip:** three cards linking to /method.html anchors — "Missing sensors are marked" · "Held-out error: X %" (from calibration file) · "Every number traces to a NASA pixel".
7. **Footer:** data citations (short), licence, AI-use note link, team names, "Not an operational fire-management tool".

C11-T2 — Remove visual noise

- Delete the floating globe console (.console-tabs, wildfire toggle, satellite-signals switch) with R1/R7.
- Max two font sizes per card; no more than one animated element on screen at a time; respect prefers-reduced-motion (already supported in design.css).

C11-T3 — Sensor timeline graphic (reused on home and method page)

- Horizontal timeline 2000 → 2027 with one lane per platform: Terra MODIS (2000-02 → ~2027-01), Aqua MODIS (2002-07 → ~2026-08), S-NPP VIIRS (2012-01-20 → delivery end 2026-11-01), NOAA-20 VIIRS (2020-01-01 →), NOAA-21 VIIRS (2024-01-17 →). For NOAA-20 and NOAA-21 these are FIRMS record start dates, not launch dates; label the lanes "in FIRMS since". Dates: FIRMS FAQ and NASA/NOAA notices (VERIFY each and cite in a tooltip).
- Outage ticks from sensor_notices.json; the overlap window used for calibration shaded; "harmonized unit: S-NPP-equivalent" label.
- ☐ Generated from sensor_notices.json + a platforms.json file; no hardcoded dates in JS.

C11-T4 — Method & validation page (/method.html) sections

1. "An observation becomes a day" (keep existing workflow visual).
2. "Two sensors, different eyes" — pixel size comparison (keep validity.js drawSensors), plus the calibration ratio per region as a bar chart.
3. "Missing sensors are marked" — sensor timeline + Park 2024 outage example from the ledger.
4. "Does harmonization work?" — C10-T1 table + scatter; C10-T2 step figure; C10-T3 burned-area correlation.
5. "Trace any number" — existing recount (keep) + calendar v2 recount.
6. "Advanced analysis" (collapsed) — merged Research Lab content (R10), HMS cross-check.
7. "Limits" — bullet list (hotspot ≠ perimeter/burned area; complete export ≠ clear pass; confidence not comparable across sensors; not for tactical use — quote FIRMS/ARSET "not recommended for use at a tactical level").

C11-T5 — New 4-step guided tour (authentic data)

1. "Pick a place" → highlights region chips (auto-selects norcal).
2. "Read 25 years at once" → highlights the heatmap, points at 2024.
3. "Was it unusual?" → highlights the August/July badge and percentile.
4. "Check the evidence" → opens Park 2024-07-25 drawer showing the S-NPP gap and raw pixels. - Replace story.js; each step uses real API data; "Skip tour" always visible.

C11-T6 — Empty, error and loading states (write every string)

State	Text
Loading calendar	Loading {region} · {years}... (skeleton rows, no spinner longer than 1 s without text)
API error	The calendar could not load. Check that the server is running, then retry. + Retry button
Outside record	No historical record is loaded for this area. Loaded areas: {list}.
Insufficient history	Fewer than 10 comparable years — no "unusual" label is given.
Degraded day	A sensor was missing or degraded on this day; the value is estimated from {source}.

C11-T7 — Reliability for judging

- Public static site (C12-T6) must work with **no server and no third-party API** except map tiles; if tiles fail, the calendar still works.
- Test in Chrome, Firefox, Safari (or WebKit via Playwright) at 1440×900 and 390×844; record results in SCORECARD.
- Lighthouse (Chrome DevTools) on the public URL: Performance ≥ 80, Accessibility ≥ 95. Record scores.

C11-T8 — Five-person usability test (counts for UX and Impact)

- Ask 5 people who never saw the app (at least one land-management, geography or environmental-science person if possible) to do three tasks without help: (1) "Find the most active month of 2024 in Northern California." (2) "Was it unusual compared with earlier years?" (3) "Find a day where a satellite was missing." Record time, success, and one quote each in `evidence/C11-T8-usability.md` (first names or initials only, with consent; no minors' names or photos).
- ☐ $\geq 4/5$ succeed on every task; fix the top problem and retest one person.

Master prompt (C11):

```
TASK C11-T<n>. Read docs/winning-plan/src/05-calendar-validation.md "C11-T<n>" and
section 3.2 (visual rules) in 03-cleanup.md. Implement the exact structure and copy.
Use only data from the APIs. Keep accessibility (keyboard, contrast, reduced motion).
Capture before/after screenshots at 1440x900 and 390x844 into
docs/winning-plan/evidence/. Run tests. Output the REPORT block.
```

C11 checklist: ☐ T1 home · ☐ T2 noise removed · ☐ T3 timeline · ☐ T4 method page · ☐ T5 tour · ☐ T6 states · ☐ T7 reliability · ☐ T8 usability test

C12 (rest) — Build, documentation, data access and reproducibility (weight 4, now 2, target 5)

C12-T1 to C12-T3 are in Gate 1 (section 5). These tasks finish the category.

C12-T4 — Open-source licence

- **Files:** `LICENSE` (new, repository root), `pyproject.toml` (`license = "Apache-2.0"`), README badge.
- **Steps:**
 1. Copy the full, unmodified Apache License 2.0 text from <https://www.apache.org/licenses/LICENSE-2.0.txt> into `LICENSE`. Do not retype it.
 2. Add NOTICE listing: "FireAtlas © 2026 the FireAtlas team (names)". Third-party data are **not** covered by the code licence; state that in README → "Data licences" and link `/data.html`.
 3. Check that every bundled third-party file (Leaflet, fonts, images in `incident-media/`) has a licence allowing redistribution. For each image record source URL, author/agency and licence in `fireatlas/static/incident-media/SOURCES.md`. Remove any image whose licence is unknown.
- **Acceptance checks:** ☐ GitHub shows "Apache-2.0" in the repo sidebar. ☐ `SOURCES.md` has a row for every file in `incident-media/`, or the file is deleted.

C12-T5 — README and PRD rewrite

- **Files:** `README.md` (≤ 120 lines), `PRD.md` (fix or move to `docs/archive/`).
- **README outline (exact headings, this order):**
 1. `# FireAtlas` + one line: A harmonized MODIS + VIIRS burning-activity calendar built on NASA FIRMS. + badges (CI, licence, live demo).
 2. `## Live demo` — public URL (C12-T6) + one screenshot of the calendar heatmap (`docs/img/calendar.png`, < 400 KB).
 3. `## What it does` — 4 bullets: multi-year calendar; harmonized S-NPP-equivalent unit with intervals; missing-sensor days marked; every number traceable to NASA pixels.
 4. `## Why it matters` — 3 sentences on sensor retirements (Terra, Aqua, S-NPP) and the need for a continuous record. Cite sources.
 5. `## Quick start` — `git clone ...`, `uv sync`, `bash scripts/launch_demo.sh`, open `http://127.0.0.1:8000`. Nothing else is needed for the preset regions.
 6. `## Data` — table: dataset, product/version, DOI or URL, licence. Link `docs/DATA.md` (C2-T6).
 7. `## Method` — 5 lines + link `docs/HARMONIZATION_METHOD.md`.
 8. `## Validation results` — a small table copied from SCORECARD with the date of the run.
 9. `## Limits` — hotspot $\neq$ burned area; not tactical; confidence not comparable; complete export $\neq$ clear pass.
 10. `## Reproduce everything` — commands to rebuild aggregates and calibration from raw FIRMS archives (C2-T2, C2-T5).
 11. `## Team, AI use and credits` — names and roles, link `docs/AI_USE.md`, NASA/FIRMS acknowledgement text.
- **Delete from README:** all mentions of training lab, PWA, synthetic demo, casebook, `--judge-demo`, Windows PowerShell launcher (unless still shipped), and duplicated install sections.
- **PRD:** fix the contradictory line (`PRD.md:130`) or move `PRD.md` to `docs/archive/PRD-2026-09.md` with a one-line note "superseded by `docs/winning-plan`".
- **Acceptance checks:** ☐ `wc -l README.md` ≤ 120 . ☐ `rg -n -i "training|pwa|judge-demo|casebook" README.md` returns nothing. ☐ A new person follows Quick start successfully (record name/initials in SCORECARD).

C12-T6 — Public static site (no server needed by judges)

- **Goal:** a stable URL that works even if nobody's laptop is on.
- **Files:** `scripts/export_static.py` (new), `.github/workflows/pages.yml` (new), `tests/test_export_static.py`.
- **Steps:**
 1. `export_static.py --out site/` copies `fireatlas/static/` and writes pre-computed JSON for every preset region and every option combination the public UI exposes: `site/api/v2/calendar/<region>/<series>-<metric>-<day>-<types>-`

<confidence>.json (use the same function as the live route; do not re-implement), plus site/api/method/*.json (calibration, availability, validation tables).

2. In JS, add one switch: `const API_MODE = document.documentElement.dataset.apiMode || "live"`. In static mode, build file paths instead of query URLs. The exporter sets `data-api-mode="static"` in the copied HTML.
3. Disable controls that need the server (custom bbox, CSV upload, recount) in static mode and show "Available in the local install".
4. Pages workflow: on push to main, run tests, run `export_static.py`, upload site/ with actions/upload-pages-artifact, deploy with actions/deploy-pages. Total size < 200 MB.
5. Also attach the full aggregates + calibration + availability files as a **GitHub Release** asset `fireatlas-data-v1.zip` with SHA-256 in the release notes.

- **Acceptance checks:**

- ☐ Public URL loads calendar for every preset region with the network panel showing no request to 127.0.0.1 and no failed request.
- ☐ Test: exporter output for `norcal` equals the live API response byte-for-byte (after JSON normalisation).

C12-T7 — Sources page and AI-use disclosure

- **Files:** `fireatlas/static/data.html` (rewrite as "Sources"), `docs/AI_USE.md` (new).
- **Sources page layout:** one card per dataset — name, provider, product/collection version, DOI/URL, licence/terms, date retrieved, file hashes, exact role in FireAtlas ("baseline input", "availability ledger", "independent check", "Tier 2 replay input"). Sections: "Harmonized record inputs" · "Validation references" · "Replay inputs (beta)" · "Map tiles and software".
- **docs/AI_USE.md template (fill truthfully):**

```
# Use of AI tools
- Tools used: <list, e.g. Cursor with <model names>>
- Used for: code drafting, test drafting, documentation drafting, planning.
- Not used for: generating any data values, images of fires, or validation results.
- Human review: every merged change was read and run by <names>.
- Space Apps rule check: we follow the Space Apps AI guidance (link) and disclose here and on the project page.
```

- **Acceptance checks:** ☐ Every dataset used in code appears on the Sources page (test: a list in `fireatlas/sources.py` drives both the page and a unit test that greps the code for dataset IDs). ☐ `AI_USE.md` linked from footer and README.

C12-T8 — Teamwork evidence (local rubric: Teamwork 5 points)

- **Steps:** each team member works on their own branch and merges through pull requests with a short review comment by another member; add `docs/TEAM.md` (name, role, what they built, with PR links). Use GitHub Issues with the task IDs from this plan (labels `tier1`, `tier2`, `submission`).
- **Acceptance checks:** ☐ ≥ 2 people have merged PRs. ☐ `docs/TEAM.md` lists roles.

C12-T9 — Release and reproducibility check

- **Steps:** tag `v1.0-submission` after the final change; run `scripts/smoke_test.sh` against the tag; rebuild aggregates from raw archives for one region on a second machine and compare SHA-256 of the output with the release asset.
- **Acceptance checks:** ☐ Hashes match (paste both). ☐ Tag exists on GitHub.

Master prompt (C12-T4...T9):

```
TASK C12-T<n>. Read docs/winning-plan/src/06-packaging-and-safety.md "C12-T<n>".
Follow the steps exactly; copy legal texts verbatim from their official source and paste
the source URL in the REPORT. Do not invent licences for images: if unknown, delete the
image and list it. Run the test suite. Output the REPORT block.
```

C12 checklist: ☐ T1 · ☐ T2 · ☐ T3 · ☐ T4 licence · ☐ T5 README · ☐ T6 static site · ☐ T7 sources + AI use · ☐ T8 teamwork · ☐ T9 release

C5, C8, C9 — Tier 1 parts (safety wording and responder-facing information)

Tier 1 does **not** build Scout. It makes sure the current product says exactly what it is and never implies operational use. This lifts C5 to 2, C8 to 3 and C9 to 3.

C8-T1 — Operational-safety wording and a banned-phrase test

- **Files:** `tests/test_safety_language.py` (new), all files in `fireatlas/static/*.html|*.js`, API text fields in `fireatlas/*.py`.
- **Banned phrases (case-insensitive) unless inside a sentence that negates them (list the allowed exact sentences in the test):** predicts, prediction, forecast, safe zone, safe area, evacuat, dispatch, authoriz, guarantee, real-time, tactical, containment line, validated (allowed only in "held-out validation" and when a validation file exists).
- **Required sentences (exact text, must appear on the home page footer and /method.html Limits):**
 - FireAtlas is a research and learning tool. It is not an operational fire-management, evacuation or flight-planning tool.

- For current fire information, warnings and orders, use the official sources listed for this region. (the list comes from `official_links` in C9-T1). Never call a non-government app or website "official".
- **Test logic:** load every static file and a set of API responses (calendar, harmonization, briefing); fail if a banned phrase appears outside the allowlist.
- **Acceptance checks:** ☐ Test passes. ☐ Screenshot of footer with the required sentence.

C9-T1 — Replace "responder briefing" with a "Season context" card and official links

- **Files:** `fireatlas/briefing.py`, `#responder-briefing` in `index.html`, `app.js`.
- **Changes:**
 1. Rename the card to **"Season context"**. Remove the label `historical-review-candidate` (`briefing.py:42-45`); instead output `record_status: comparable` (same product versions and all sources available), `mixed-versions`, `degraded-days` (k days listed), or `insufficient-history`.
 2. Content: selected period's value vs climatology (from C1-T3), season start shift (C1-T4), degraded days, and "What to check next" = links only.
 3. **Official links panel** per preset region, stored in `fireatlas/regions.py` as `official_links: [{label, url, kind}]` where `kind` $\in$ {agency, alerts, incidents, airspace}. Seed (VERIFY every URL loads before commit):
 - `norcal / california`: CAL FIRE incidents (<https://www.fire.ca.gov/incidents>), InciWeb (<https://inciweb.wildfire.gov>), FAA TFR list (<https://tfr.faa.gov>).
 - `punjab-haryana`: VERIFY official state / national sources (e.g., Punjab Remote Sensing Centre, Indian Agricultural Research Institute CREAMS residue-burning bulletins); if none can be verified, show only NASA FIRMS and write "No verified official local source listed".
 - `rondonia`: INPE Queimadas (<https://terrabrasilis.dpi.inpe.br/queimadas/portal/>) — VERIFY.
 - `top-end-au`: North Australia and Rangelands Fire Information (<https://firenorth.org.au>) and NT Bushfires — VERIFY.
 4. Remove any "context not integrated" placeholder text (`briefing.py:76-81`); omit the section instead.
- **Acceptance checks:** ☐ API returns `record_status` values in tests for all four states. ☐ Each link was loaded (HTTP 200/301) — paste the curl results. ☐ No link is labelled "official" unless it is a government or agency site.

C5-T1 — Scope statement for aerial observation (no Scout yet)

- Add to `/method.html` → "Where this could go" (3 short paragraphs, no promises):
 1. The harmonized calendar tells *when and where fires usually happen*; that helps plan **when** observation assets (satellite tasking, aerial surveys) are most useful.
 2. Any drone or aircraft use near wildfires is governed by the incident's aviation organisation and airspace rules (US: FAA TFRs under 14 CFR 91.137; NWCG Fire Traffic Area procedures in the NWCG Standards for Airspace Coordination / PMS 515 — VERIFY document titles). FireAtlas does not plan flights.
 3. The "Fire replay (beta)" extension (if shipped) studies, on a past fire, which areas were most uncertain — as a research question, not an operational tool.
- **Acceptance checks:** ☐ Text present; C8-T1 test passes on it; the citation URLs are on the Sources page.

Master prompt (C5/C8/C9 Tier 1):

TASK <C8-T1|C9-T1|C5-T1>. Read docs/winning-plan/src/06-packaging-and-safety.md for the task. Implement the exact wording given. For every URL, run `curl -sI <url>` and paste the status line; do not add a URL you could not load. Do not add any feature that suggests operational use. Run the test suite (including tests/test_safety_language.py). Output the REPORT block.

Checklist: ☐ C8-T1 · ☐ C9-T1 · ☐ C5-T1

7. Tier 2 — FireAtlas Scout: Park Fire 2024 uncertainty replay (beta)

Start condition: every Tier 1 acceptance check for C1, C2 and C10-T1...T7 is `DONE` in SCORECARD. If the hackathon is less than 10 days away (Branch A) or less than 14 hours remain (Branch B) and Tier 1 is not done, mark all of section 7 `DROPPED (time)` and polish Tier 1 instead. A strong Tier 1 beats a weak Tier 1 plus a weak Tier 2.

What Scout is (one sentence, use it everywhere): *A replay of a past fire that shows, hour by hour, which areas the fire-spread scenarios disagreed about most—and how new observations reduced that uncertainty.*

What Scout is not: a forecast, a flight planner, an evacuation tool, or anything usable on an active fire.

Case: Park Fire, Butte/Tehama counties, California, ignited 24 July 2024 (VERIFY ignition time on the CAL FIRE incident page and record it). Replay window: ignition → 31 July 2024, time step 6 hours. This case is chosen because (a) S-NPP VIIRS was out 24–28 July, so the replay shows how the system copes with a missing sensor using MODIS and NOAA-20 VIIRS; (b) it is already the flagship example in the bundle.

7.0 Shared foundations

S-T0 — Replay clock and leakage guard (C10 Tier 2)

- **Files:** `fireatlas/replay.py` (new), `tests/test_replay.py`.
- **Design:**
 1. Every replay input is a record with `valid_at` (time the value describes, UTC) and `available_at` (earliest time a person could have had it, UTC). Rules for `available_at`: FIRMS NRT detections = acquisition time + 3 h (FIRMS NRT latency target; VERIFY on the FIRMS FAQ and record); standard-product detections are **not allowed** in replay (they arrived months later) — use the NRT archive files for July 2024 (VERIFY that FIRMS archive download offers NRT for that date; if not, use standard detections with `available_at` = acquisition + 3 h and write the limitation on the page); HRRR analysis = valid time + 1 h; aerial perimeters = file timestamp; terrain and fuels = static (`available_at` = 1970-01-01), with the fuels product year ≤ 2024.
 2. `class ReplayClock(now_utc)` with inputs(`kind`) → list that returns only records with `available_at` ≤ `now_utc`. All model code receives inputs **only** through the clock.
 3. Every output file stores `as_of_utc` and the list of input IDs used.
 4. **Active-fire block:** the replay endpoint refuses any event whose last detection is within 30 days of the server's current date (HTTP 403, message "Replay is only for past fires"). Test it.
- **Acceptance checks:**
 - ◻ Test: a record with `available_at` one second after `now_utc` is never returned.
 - ◻ Test: output files list only input IDs with `available_at` ≤ `as_of_utc` (property test over 100 random clocks).
 - ◻ Test: active-fire block works.

C3 — Terrain, fuels, vegetation and dated weather (weight 8, now 1, target 4)

C3-T1 — Terrain

- **Data:** NASADEM 1 arc-second (≈30 m), NASA LP DAAC, Earthdata Login (VERIFY product short name `NASADEM_HGT v001`). Fallback: Copernicus GLO-30 DEM (licence attribution required).
- **Files:** `fireatlas/terrain.py`, `data/replay/park-2024/dem.tif` (not committed; add to Release asset), `tests/test_terrain.py`.
- **Steps:** download tiles covering `bbox -122.3, 39.6, -121.4, 40.4` (VERIFY it covers the final Park perimeter plus 10 km); mosaic and reproject to EPSG:32610 (UTM 10N), 30 m; compute slope (degrees), aspect (degrees from north), TPI (elevation minus mean in 300 m radius). Use GDAL (`gdaldem slope/aspect`) or numpy; record the commands.
- **Acceptance checks:** ◻ Unit test on a synthetic tilted plane gives the known slope/aspect. ◻ Summary stats written to SCORECARD.

C3-T2 — Fuels

- **Data:** LANDFIRE 13 Anderson / 40 Scott-Burgan fuel models (FBFM40), product version dated **before** 24 Jul 2024 (VERIFY: LF 2022 "LF 2.3.0" or LF2023 "LF 2.4.0" release date). Download via the LANDFIRE Product Service (LFPS) or the map viewer.
- **Steps:** clip to `bbox`, resample (nearest) to the 30 m DEM grid, map codes to the Scott & Burgan parameter table used by the spread model; codes 91–99 (non-burnable) set to non-burnable.
- **Acceptance checks:** ◻ Code histogram in SCORECARD. ◻ Test: non-burnable codes never ignite in the spread model.

C3-T3 — Dated weather

- **Data:** NOAA HRRR analysis (f00) hourly, from the NOAA Open Data Dissemination bucket `noaa-hrrr-bdp-pds` on AWS (VERIFY bucket name and file naming). Variables: 10 m U/V wind, 2 m temperature, 2 m RH. Fallback: gridMET daily (lower resolution; label it).
- **Steps:** for each hour in the replay window, extract the variables on the replay grid (bilinear), store as `data/replay/park-2024/weather/<valid_utc>.npz` with `available_at` = `valid` + 1 h. Compute dead fuel moisture (1-h, 10-h) with a documented simple method (e.g., Fosberg/Nelson table — VERIFY source and cite).
- **Acceptance checks:** ◻ Freshness panel (C7-T3) shows the age of weather at each step. ◻ Test: weather interpolation reproduces a constant field exactly.

C7 — Spread scenarios, uncertainty and input freshness (weight 10, now 1, target 4)

C7-T1 — Spread model

- **Preferred:** `pyretechnics` (Rothermel surface spread + elliptical growth; VERIFY the package exists on PyPI, its licence is OSI-compatible, and its API) — pin the version.
- **Fallback (only if preferred fails in < 4 h of effort):** a cellular automaton on the 30 m grid where the spread probability to a neighbour = $p_0 \times f_{\text{fuel}} \times \exp(a \cdot \text{wind_along}) \times \exp(b \cdot \text{slope_along})$, with p_0 , a , b from a cited paper (e.g., Alexandridis et al. 2008 — VERIFY values). Label it on screen as "**simplified, uncalibrated spread model**".
- **Files:** `fireatlas/spread.py`, `tests/test_spread.py`.
- **Acceptance checks:** ☐ Test: no wind and flat terrain → roughly circular spread (aspect ratio < 1.2). ☐ Test: wind from west → spread further east. ☐ Test: non-burnable cells never burn.

C7-T2 — Ensemble (50 members)

- **Perturbations per member (seeded, seed = 20261114 + member):** wind speed $\times U(0.8, 1.2)$; wind direction $+ N(0^\circ, 15^\circ)$; dead fuel moisture $\times U(0.85, 1.15)$; ignition = observed hotspot cells as of `as_of_utc` (from the clock) with each cell kept with probability 0.9.
- **Output per step:** burn-probability grid $p(x)$ for horizons +6 h, +12 h, +24 h = fraction of members that burned cell x .
- **Acceptance checks:** ☐ Same seed → identical output (test). ☐ Runtime per step < 60 s on the dev laptop (record).

C7-T3 — Freshness and uncertainty display

- For each step show a **freshness table**: input · valid time · available time · age at `as_of` (e.g., "NOAA-20 VIIRS detections · 25 Jul 09:48 UTC · 12:48 UTC · 2 h 12 min old"). Rows: MODIS, S-NPP (shows "missing — NASA notice"), NOAA-20, weather, fuels, terrain, aerial perimeter.
- Map layers: $p(x)$ as a sequential colour ramp with a legend "share of 50 scenarios that burned this cell"; the **uncertainty band** = cells with $0.1 < p < 0.9$ outlined.
- **Acceptance checks:** ☐ Screenshot at three clocks. ☐ Legend text exactly as above.

C4 — Explainable ranking of observation sectors (weight 14, now 0, target 4)

Question answered: *Which parts of the fire edge were the scenarios most unsure about at this time, and why?* This is where an extra observation would have reduced uncertainty the most.

C4-T1 — Sectors

- Divide the area within 10 km of the current observed fire extent into 2 km $\times$ 2 km sectors on the UTM grid (`fireatlas/sectors.py`). Keep only sectors with at least one cell where $0.05 < p < 0.95$ at +12 h.

C4-T2 — Score and components (all in [0, 1])

- U (uncertainty) = mean over sector cells of binary entropy $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ at +12 h.
- E (exposure) = sector building count / max building count across sectors. Data: Microsoft US Building Footprints (ODbL; VERIFY) or OpenStreetMap buildings (ODbL). Record the data date; buildings built after 2024 must not be used.
- S (staleness) = $\min(1, \text{hours since the last satellite detection-capable overpass covering the sector (from the CMR availability ledger, C2-T3)} / 12)$.
- V (terrain visibility, illustrative) = fraction of sector cells visible from a point 120 m above the sector's highest cell (viewshed on the DEM; `gda_viewshed`). It shows how much of the sector a single elevated viewpoint could see; it is **not** a flight altitude recommendation.
- **Score** = $U \times (0.5 + 0.5E) \times (0.5 + 0.5S) \times V$. Write the formula on screen.
- **Acceptance checks:** ☐ Unit tests for each component on hand-made grids. ☐ Test: a sector with $p = 0$ or 1 everywhere has $U = 0$ and score 0.

C4-T3 — Explanation UI

- Right panel "Most uncertain sectors at 25 Jul 18:00 UTC": ranked list (top 5) with a stacked bar showing the four components and a one-line reason generated from the largest components, e.g., "Scenarios split 48/52 here; 120 buildings; last satellite look 9 h ago."
- Clicking a sector highlights it and shows the member-by-member burn outcome (small multiples of 10 members).
- **Ablation note** (below the list): rank correlation between the full score and U alone — shows what E , S and V change.
- **Acceptance checks:** ☐ Every number in the reason line comes from the component values (test). ☐ No text uses the words "fly", "deploy", "send" or "dispatch" (safety test from C8-T1 covers the page).

C4-T4 — Does the ranking mean anything? (retrospective check)

- For each step, compare the top-5 sectors with where the fire actually grew in the next 12 h (from the next observed perimeter): **hit rate** = share of top-5 sectors that intersect new growth; compare with (a) 5 random edge sectors (1,000 draws; report mean and 95 % range) and (b) 5 sectors with the highest p (most likely to burn).
- Report honestly on the page, including if the ranking is no better than random.
- **Acceptance checks:** ☐ Table in SCORECARD with hit rates per step and baseline ranges.

C6 — Dated aerial observations and state update (weight 8, now 0, target 4)

C6-T1 — Observation ingest

- **Data (in order of preference, VERIFY each for the Park Fire):** (1) NASA FEDS (Fire Event Data Suite) VIIRS-derived 12-hourly fire perimeters (NASA Earth Information System; VERIFY access location and licence); (2) NIFC / FIRIS airborne infrared perimeters from the NIFC Open Data site (WFIGS or the interagency perimeter history). Record each perimeter's `observed_at` and `available_at`.
- **Files:** `fireatlas/observations_aerial.py`, `data/replay/park-2024/perimeters/*.geojson`, `tests/test_observations_aerial.py`.
- **Rules:** perimeters without a timestamp are rejected; geometry must be valid (fix with `buffer(0)` and log it); reproject to UTM 10N.
- **Acceptance checks:** ☐ ≥ 6 dated perimeters in the replay window, or a written reason why not. ☐ Ingest test with a tiny GeoJSON.

C6-T2 — State update (assimilation)

- **Files:** `fireatlas/assimilate.py`, `tests/test_assimilate.py`.
- **Method:** when a new perimeter P arrives at time t : for each member k , compute Jaccard J_k between its burned area at t and P . Weight $w_k \propto \exp(-(1 - J_k)/0.1)$, normalise. Resample 50 members by systematic resampling using the weights. Re-seed all members from P (burned = inside P) while keeping each member's perturbation parameters. Record effective sample size $1/\sum w^2$ per update.
- **Acceptance checks:** ☐ Test: identical member and perimeter $\rightarrow J = 1$ and highest weight. ☐ Test: weights sum to 1. ☐ ESS plotted per update.

C6-T3 — Replay metrics (C10 Tier 2)

- For each step with a next perimeter: Jaccard of the $p \geq 0.5$ footprint vs next perimeter; Brier score of p vs observed burned/unburned cells within 10 km. Compare three versions: **persistence** (perimeter does not grow), **ensemble without assimilation**, **ensemble with assimilation**.
- **Acceptance checks:** ☐ Table and line chart on `/replay.html` $\rightarrow$ "How well did it do?" with all three. ☐ Numbers in SCORECARD.

C5 (Tier 2) — Aerial-observation feasibility framing and airspace/command boundaries (target 4)

C5-T2 — Boundaries panel (text and diagram, not map overlays)

- On `/replay.html` add a collapsible "Who controls the airspace" panel:
 1. Over a US wildfire, the FAA usually issues a Temporary Flight Restriction under 14 CFR 91.137; unauthorised drones are prohibited, and drone incursions have grounded firefighting aircraft (cite the US Forest Service / NIFC "If you fly, we can't" material — VERIFY the page).
 2. Incident aviation is coordinated by the incident's air operations organisation using NWCG procedures (Fire Traffic Area; cite the NWCG document — VERIFY title and number).
 3. FireAtlas outputs are **research questions** ("where was uncertainty highest?") that an authorised team *could* consider; the tool gives no routes, altitudes or targets.
- A simple static SVG diagram: "Satellites (FIRMS) $\rightarrow$ analysts $\rightarrow$ incident air operations (decides) $\rightarrow$ authorised aircraft/UAS" with the decision box clearly owned by the incident.
- **Acceptance checks:** ☐ No coordinates, rings or lines are drawn on the map for airspace. ☐ Citations on the Sources page.

C5-T3 — Feasibility numbers (illustrative, clearly labelled)

- For the top-5 sectors, show only: sector area (km²), terrain visibility V , and "last satellite look" age. Do **not** compute flight times, battery, routes or altitudes.

C8 (Tier 2) — Containment boundaries and operational safety (target 4)

C8-T2 — Replay safety design

- Persistent top banner on `/replay.html` (not dismissible): Replay of a past fire (Park Fire, July 2024). Scenarios, not forecasts. Not for operational use.
- The observed perimeter is drawn as a solid line labelled "Observed perimeter (source, time)"; scenario output is only a probability raster. **Never** draw or label containment lines, control lines, dozer lines, "safe" areas or evacuation zones.
- The active-fire block from S-T0.
- **Acceptance checks:** ☐ Banner present on every replay screenshot. ☐ C8-T1 banned-phrase test covers `replay.html/js` and replay API text.

C9 (Tier 2) — Responder workflow, resident information, official-alert separation (target 4)

C9-T2 — After-action review workflow

- Replay controls: play/pause, step ± 6 h, time slider, "jump to next observation". At each step the right panel shows: what was known (freshness table), what scenarios said, what happened next (revealed only after pressing "Reveal outcome" — this keeps the leakage honest for viewers too).
- "Export review" button → Markdown/PDF with the step table, metrics and all source citations.

C9-T3 — Resident separation

- A small "Live in this area?" box that contains **only** the official links from `regions.py` (C9-T1) and the sentence `This replay is about a past fire. For current conditions, use official sources.` No FireAtlas data inside the box.

7.x Replay page layout (`/replay.html`)

1. Banner (C8-T2).
2. Title `Park Fire 2024 – where was the fire's future most uncertain?` + one-line explanation.
3. Main: Leaflet map (satellite detections by sensor colour, observed perimeter, `p(x)` raster, sector outlines with rank numbers). Base map: a plain terrain/hillshade tile layer; the hillshade can be generated from the DEM so no third-party globe is needed.
4. Right panel tabs: `Now` (freshness + top sectors), `Why` (component bars, formula), `Outcome` (hidden until revealed), `Accuracy` (C6-T3 metrics).
5. Bottom: time slider with ticks for each observation (satellite overpasses coloured by sensor, perimeters as white ticks, S-NPP outage shaded).
6. Footer: methods summary, limits, citations, "Who controls the airspace" (C5-T2).

Master prompt (Tier 2, one task at a time):

```
TASK <S-T0|C3-Tn|C7-Tn|C4-Tn|C6-Tn|C5-Tn|C8-T2|C9-Tn>. Read docs/winning-plan/src/07-scout-replay.md for that task. Tier 2 may start only if SCORECARD shows all Tier 1 checks DONE; if not, stop and say so. All inputs must pass through fireatlas/replay.py ReplayClock. Every dataset marked VERIFY must be confirmed from its official page (paste URL and version) before use; if it cannot be confirmed, mark BLOCKED. Report metrics exactly as computed, including bad results. Never output routes, waypoints, altitudes or containment lines. Add tests. Run the full suite. Output the REPORT block.
```

Tier 2 checklist: ☐ S-T0 · ☐ C3-T1 · ☐ C3-T2 · ☐ C3-T3 · ☐ C7-T1 · ☐ C7-T2 · ☐ C7-T3 · ☐ C4-T1 · ☐ C4-T2 · ☐ C4-T3 · ☐ C4-T4 · ☐ C6-T1 · ☐ C6-T2 · ☐ C6-T3 · ☐ C5-T2 · ☐ C5-T3 · ☐ C8-T2 · ☐ C9-T2 · ☐ C9-T3 · ☐ `/replay.html` linked in nav only after all above are DONE

8. Submission package

Everything here is judged directly. Draft it early (week 6 in Branch A), finalise it at the event.

SUB-T1 — Project page (Space Apps site)

Fill every field; the local rubric gives 1 point just for a complete page. Field names below follow the 2025 project-page form — **VERIFY against the 2026 form** and adapt the mapping, not the content.

Field	What to write (exact guidance)
Project name	FireAtlas — one fire calendar across MODIS and VIIRS
Challenge	Harmonization of MODIS and VIIRS Hot Spots (exact 2026 title from G0-T3)
High-level summary (≤ 150 words)	Problem (sensor handover, retirements) → what FireAtlas does (harmonized calendar, missing-sensor days, uncertainty, trace to pixel) → one result with your measured numbers → who it helps. Use the winning story (2.4).
Link to final project	Public static site URL (C12-T6)
Link to repository	GitHub URL (public)
Presentation / video	30-second video link (SUB-T2) or 7-slide deck (SUB-T3)
Detailed description	Sections: What it does · How it works (method in 6 bullets) · Validation (tables from SCORECARD with run dates) · Limits · Future work (Tier 2 replay, more regions, NOAA-21)
Use of NASA data	Name every NASA product with version and DOI/URL: FIRMS MODIS C6.1 (MCD14ML), FIRMS VIIRS S-NPP (VNP14IMGML), NOAA-20 (VJ114IMGML), CMR granule metadata (MOD14/MYD14/VNP14IMG/VJ114IMG), MCD64A1 C6.1, NASADEM (if Tier 2), FEDS (if Tier 2). Say what each is used for.
Other data	NOAA HMS, LANDFIRE, HRRR, CAL FIRE incidents, building footprints — with licences.
Use of AI	Copy the summary of docs/AI_USE.md.
References	All citations (C2-T6 list, papers used in C7).
Prior work disclosure (Branch A only)	Parts of this project (data pipeline and prototype interface) were developed between 27 Sep and 13 Nov 2026, before the hackathon, with the Local Lead's written approval on <date>. During the hackathon we built: <list with commit range>. Use the exact ruling wording.
Team	Names, roles, countries; no personal contact details.

- **Acceptance checks:** ☐ Every field filled. ☐ Every number matches SCORECARD (same run date). ☐ Links open in a private browser window.

SUB-T2 — 30-second video (storyboard, exact timing)

Record at 1920×1080, 30 fps, screen capture of the public site. Voice-over (VO) ≤ 75 words total; burn-in captions; no music with lyrics; end card ≥ 2 s.

Time	Screen	VO (exact draft; replace bracketed values with SCORECARD numbers)
0:00–0:04	Sensor timeline animating; Terra, Aqua, S-NPP lanes end	"NASA's fire satellites are retiring. Twenty-five years of fire records are split across sensors that see fire differently."
0:04–0:11	Heatmap for Northern California, 2003–2025, scrolling	"FireAtlas joins MODIS and VIIRS into one burning-activity calendar—in a single unit, with uncertainty."
0:11–0:17	Hover the 2024 July row; hatched outage days; tooltip	"Missing-sensor days are marked, not hidden—like the Park Fire, when Suomi NPP went dark."
0:17–0:23	Method page: held-out error table, scatter	"On years we held out, it predicts VIIRS activity within [X] percent."
0:23–0:27	Day drawer → raw NASA rows	"Every number traces back to the original NASA pixel."
0:27–0:30	End card: logo, URL, challenge name, "Data: NASA FIRMS"	(silence)

- **Checks:** ☐ ≤ 30.0 s (check file duration with `ffprobe`). ☐ Captions readable on a phone. ☐ No synthetic data on screen. ☐ Shown to 2 people who have not seen the project; both can say what it does.

SUB-T3 — 7 slides (if slides instead of or in addition to video)

1. **Title** — name, one-line promise, challenge name, team.
2. **Problem** — sensor timeline graphic; "A single sensor can't tell 25 years of fire history anymore."
3. **What we built** — screenshot of the heatmap with 3 callouts (harmonized unit · degraded days · trace to pixel).
4. **How it works** — 5-box flow: FIRMS archives → 1 km cell-days → availability ledger → overlap calibration → calendar + intervals.
5. **Does it work?** — held-out error table (vs no harmonization and single-ratio baselines) + burned-area correlation; one sentence of limits.
6. **Who it helps** — land manager question answered in 60 s (usability test result); second region example (crop or savanna burning).
7. **What's next + credits** — Tier 2 replay (if built: one screenshot, "beta"), NOAA-21 continuity, data credits, AI use, links/QR code.

Rules: ≥ 24 pt font, ≤ 25 words per slide except slide 5 table, every chart captioned with source.

SUB-T4 — Live demo run sheet (for local judging, 3-5 minutes)

- 1. Open the public URL (not localhost) in a fresh browser profile; have the local server running as backup and a 60 s screen recording as a last resort.
 - 2. Say the one question (2.4). Pick Northern California chip.
 - 3. Point at the heatmap; hover a 2024 hatched day; open the day drawer; show raw rows.
 - 4. Switch to Punjab-Haryana to show crop-burning seasons (global relevance).
 - 5. Open Method → "Does harmonization work?"; read one number.
 - 6. (If Tier 2 done) Open Fire replay: step two times, reveal outcome, show accuracy vs persistence.
 - 7. Close with limits and the challenge name.
- **Checks:** ☐ Rehearsed 3 times with a timer; each run ≤ 5 min. ☐ Offline fallback tested (Wi-Fi off → local server + static site folder).

Master prompt (submission drafting):

```
TASK SUB-T<n>. Read docs/winning-plan/src/08-submission-tracker-timeline.md "SUB-T<n>"
and docs/winning-plan/SCORECARD.md. Draft the text exactly to the structure given.
Take every number, date and URL from SCORECARD or the repository; if a value is missing,
write [MISSING: what] instead of guessing. No superlatives ("revolutionary",
"first-ever", "real-time") and no operational claims. Output the draft and a list of
every number used with its SCORECARD row.
```

Submission checklist: ☐ SUB-T1 · ☐ SUB-T2 · ☐ SUB-T3 · ☐ SUB-T4 · ☐ submitted before 15 Nov 18:00 local (6 h buffer before the 23:59 deadline) · ☐ confirmation screenshot saved

9. The live accountability tracker

`docs/winning-plan/SCORECARD.md` is the single source of truth for progress. The PDF is the plan; the SCORECARD is the state.

9.1 Update rules

1. After **every** task, the executor's REPORT block says which SCORECARD rows change. You (or the Auditor) apply them.
2. A task row may be set to `DONE` only with evidence (commit hash + test line + screenshot/JSON path).
3. **Category scores are never set by the executor.** They are recomputed by the Auditor (9.2) at least every 2 days and after each block of section 0.1.
4. Every change adds one line to the SCORECARD **Changelog**: `YYYY-MM-DD HH:MM · who · what changed · evidence`.
5. Numbers in "Ground truth & measured results" are replaced only by values from a command run in the same session; keep the old value in the changelog.

9.2 Auditor master prompt (paste into a strong LLM with repository access)

```
You are the FireAtlas AUDITOR. You do not write features. You measure.
Inputs: the repository, docs/winning-plan/SCORECARD.md, and the scoring rules in
docs/winning-plan/src/02-gates-and-scoring.md section 2.1 (scores 0-5, weights, hard caps).

Do, in order:
1. git status; git log -5 --oneline; git log origin/main -1. Note uncommitted work.
2. uv run python -m unittest discover -s tests -> paste the final line.
3. For each task marked DONE in SCORECARD: check that its evidence exists (file path,
   commit, test name). If any is missing, change the status to IN PROGRESS and say why.
4. Start the server; request /, /method.html, /data.html and the calendar v2 API for each
   preset region; record status codes and response times.
5. For each category C1..C12: choose the score 0-5 using the evidence table in 2.1; then
   apply every hard cap (synthetic-only <=2; not pushed <=2; no method+test <=2;
   forecast without held-out evaluation <=3; operational instruction => 0).
   Write one line of evidence per category with file:line or command output.
6. Compute points = weight x score / 5 for each category and the total. Show the arithmetic.
7. Estimate the local-rubric scores (section 2.2) with one sentence of justification each.
8. Output:
   a) the updated "Category scores" and "Local rubric" tables for SCORECARD,
   b) "Change since last audit" (per category: old -> new, reason),
   c) "Top 3 blockers" and "Next 3 tasks" (task IDs from the plan),
   d) a changelog line.
Rules: never raise a score without new evidence; when unsure, choose the lower score;
never invent measurements.
```

9.3 Daily improvement report (what you ask for each day)

Prompt: Using the Auditor results and SCORECARD, write today's improvement report: score now vs yesterday vs target (both systems), tasks finished today with evidence, tasks slipped and why, risk level for the deadline (green/amber/red with reason), and tomorrow's 3 tasks.

Risk level rule: **red** if the Tier 1 freeze date will be missed at the current pace (remaining Tier 1 tasks ÷ tasks done per day > days until freeze); **amber** if within 2 days of that; otherwise **green**.

10. Timelines

10.1 Branch A — reuse allowed (29 Sep → 15 Nov 2026)

Week	Dates	Work (task IDs)	Exit check
1	29 Sep – 4 Oct	G0-T1 email; Gate 1 (C12-T1...T3); cleanup R1-R17	Repo public, CI green, cleanup checklist done
2	5 – 11 Oct	C2-T1 regions + downloads (start downloads day 1; they are slow); C2-T2 aggregates; C2-T3 availability ledger	Aggregates for norcal + 1 other region
3	12 – 18 Oct	C2-T4 filters/versions; C2-T5 calibration + LOYO; C2-T6 docs; C1-T1 API v2	Calibration JSON for all regions; API contract tests
4	19 – 25 Oct	C1-T2...T5 (heatmap, climatology, seasons, area picker); C10-T1, C10-T2	Heatmap screenshot; Gate 0 answer (default Branch B if none by 25 Oct)
5	26 Oct – 1 Nov	28 Oct: G0-T3 (re-plan if needed); C1-T6...T8; C10-T3...T7; C11-T1...T4	Method page shows validation
6	2 – 8 Nov	C11-T5...T8 (tour, states, reliability, usability test); C12-T4...T8; C8-T1, C9-T1, C5-T1; submission drafts	Tier 1 freeze 8 Nov ; Auditor run; target ≥ 50/100
7	9 – 13 Nov	If Tier 1 all DONE: S-T0, C3-T1...T3, C7-T1...T3, C4-T1...T3 (core replay). Otherwise: Tier 1 fixes only. Record video draft; rehearse	Replay beta or DROPPED; video draft ≤ 30 s
Event	14 – 15 Nov	Final data refresh check; C4-T4, C6 if time; SUB-T1...T4; C12-T9 tag	Submitted by 15 Nov 18:00 local

10.2 Branch B — rebuild at the event (36 working hours, Tier 1 only)

Before 14 Nov you may prepare **only** non-code material (if the ruling allows): this plan, data-source lists, and download instructions. Confirm whether pre-downloading public datasets is allowed; if not, start downloads at hour 0.

Hours	Work
0–2	New repo, licence, README skeleton, CI; start FIRMS archive downloads for 2 regions (norcal + one contrasting region)
2–8	Ingest + 1 km cell-day aggregates (C2-T2 simplified: vegetation filter, all confidences); availability ledger from CMR for the 2 regions (C2-T3)
8–14	Calibration r(m) + LOYO vs no-harmonization baseline (C2-T5); method doc
14–22	Calendar API + heatmap + climatology badges + day drawer with raw rows (C1-T1...T3, T6)
22–26	Method page with sensor timeline, validation table, limits (C11-T3, T4 parts; C10-T1)
26–29	Static site deploy (C12-T6), safety wording (C8-T1), sources + AI use (C12-T7)
29–33	Usability test with 3 people at the venue; fix top problem; video + slides
33–35	Project page (SUB-T1), rehearsal
35–36	Submit; buffer

Take regular breaks and sleep at least one block; tired teams make validity mistakes that judges notice.

Appendix A — Schemas (implement these field names exactly)

All JSON files are UTF-8, dates are ISO 8601 (YYYY-MM-DD, times with Z), numbers are plain JSON numbers (no strings), missing values are null. Every file has `schema` and `method_version` at the top level. Add a JSON Schema file for each under `fireatlas/schemas/` and validate in tests (a small hand-written validator is fine; do not add a large dependency).

A.1 `fireatlas-daily-aggregates-v1` (`fireatlas/samples/aggregates/<region>.json.gz`)

```
{
  "schema": "fireatlas-daily-aggregates-v1",
  "method_version": "cell-days-1km-epsg6933-v2",
  "region": {"id": "norcal", "bbox": [-122.2, 38.8, -120.0, 41.0]},
  "generated_utc": "2026-10-10T12:00:00Z",
  "inputs": [{"bundle": "nasa_firms_norcal_2000_2025.zip", "sha256": "...", "source_id": "MODIS_SP"}],
  "series": ["MODIS_SP", "VIIRS_SNPP_SP", "VIIRS_N0AA20_SP"],
  "variants": ["utc|veg|all", "utc|veg|no-low", "utc|all|all", "utc|all|no-low",
    "solar|veg|all", "solar|veg|no-low", "solar|all|all", "solar|all|no-low"],
  "complete_months": {"MODIS_SP": ["2000-11", "..."], "VIIRS_SNPP_SP": ["2012-02", "..."]},
  "days": [
    {"date": "2024-07-25",
      "v": {"utc|veg|all": {
        "MODIS_SP": {"cells": 415, "raw": 603, "frp_sum": 12345.6},
        "VIIRS_SNPP_SP": {"cells": 0, "raw": 0, "frp_sum": 0.0},
        "VIIRS_N0AA20_SP": {"cells": 0, "raw": 0, "frp_sum": 0.0},
        "union_modis_snpp": 415, "intersection_modis_snpp": 0}}}
  ]
}
```

Rules: a day appears only if at least one series has a complete export for that month; the example numbers other than the Park MODIS values (603 / 415, measured) are placeholders — tests must use real computed values. Days with zero everywhere may be omitted **only** if `complete_months` covers them (readers then treat them as observed zero).

A.2 `fireatlas-calibration-v1` (`fireatlas/samples/calibration/<region>.json`)

```
{
  "schema": "fireatlas-calibration-v1",
  "method_version": "overlap-ratio-moy-v1",
  "region": "norcal",
  "pair": {"reference": "VIIRS_SNPP_SP", "scaled": "MODIS_SP"},
  "variant": "utc|veg|all",
  "overlap": {"start": "2012-02", "end": "2025-12", "years": [2012, 2013], "excluded_days": 37},
  "bootstrap": {"replicates": 1000, "seed": 20261114},
  "ratio_by_month": [
    {"month": 1, "r": 0.0, "r_lo": 0.0, "r_hi": 0.0, "sum_modis": 0, "sum_viirs": 0, "basis": "month-of-year"}
  ],
  "r_all": {"r": 0.0, "r_lo": 0.0, "r_hi": 0.0},
  "stability": {"spearman_rho": 0.0, "p_value": 0.0, "permutations": 5000, "ratio_trend_flag": false},
  "loyo": {
    "month_of_year": {"median_abs_log_error": 0.0, "annual_abs_pct_error": 0.0, "interval_coverage_pct": 0.0},
    "single_ratio": {"median_abs_log_error": 0.0, "annual_abs_pct_error": 0.0, "interval_coverage_pct": null},
    "no_harmonization": {"median_abs_log_error": 0.0, "annual_abs_pct_error": 0.0, "interval_coverage_pct": null},
    "selected_model": "month_of_year",
    "per_year": [{"year": 2013, "month_of_year": 0.0, "single_ratio": 0.0, "no_harmonization": 0.0}]
  },
  "successor": {"pair": {"reference": "VIIRS_SNPP_SP", "scaled": "VIIRS_N0AA20_SP"}, "ratio_by_month": [], "r_all": {}}
}
```

0.0 values above are placeholders showing type only. `basis` ∈ {month-of-year, annual-fallback}. `selected_model` is the model with the lowest LOYO median absolute log error; the UI uses it.

A.3 fireatlas-calendar-v2 (GET /api/v2/calendar)

```
{
  "schema": "fireatlas-calendar-v2",
  "meta": {
    "region": "norcal", "bbox": [-122.2, 38.8, -120.0, 41.0],
    "series": "harmonized", "unit": "S-NPP-equivalent cell-days", "metric": "cell_days",
    "day": "utc", "types": "veg", "confidence": "all",
    "baseline": {"requested": "2003-2022", "years_used": [2003, 2004], "excluded": [{"year": 2012, "reason": "mixed-collection"}]},
    "calibration_id": "norcal:overlap-ratio-moy-v1:utc|veg|all",
    "method_versions": {"aggregates": "cell-days-1km-epsg6933-v2", "calibration": "overlap-ratio-moy-v1", "availability": "cmr-ledger-v1"},
    "data_class": "authentic", "generated_utc": "..."
  },
  "years": [
    {"year": 2024,
      "days": [{"date": "2024-07-25", "value": 0.0, "low": 0.0, "high": 0.0,
        "estimate_type": "scaled", "source_used": "MODIS_SP", "quality": "degraded",
        "availability": {"MODIS_SP": "available", "VIIRS_SNPP_SP": "missing", "VIIRS_N0AA20_SP": "available"},
        "notice_ids": ["snpp-2024-07-processing-stop"]}],
      "monthly": [{"month": 7, "value": 0.0, "low": 0.0, "high": 0.0,
        "climatology": {"p10": 0.0, "p50": 0.0, "p90": 0.0, "n_years": 20},
        "percentile_rank": 0.0, "rank": 1, "n_ranked": 21, "flag": "unusually-high",
        "degraded_days": 5}],
      "season": {"status": "ok", "start": "2024-06-30", "peak": "2024-07-28", "end": "2024-09-20",
        "length_days": 83, "start_shift_days": -12,
        "critical_periods": [{"start": "2024-07-24", "end": "2024-08-06", "peak_value": 0.0}]}
    ],
    "provenance": [{"source_id": "MODIS_SP", "bundle_sha256": "...", "doi_or_url": "..."}],
    "limits": ["Detections show where satellites recorded heat; they are not burned area.",
      "A zero on an available day means no detections were recorded, not proof of no fire."]
  ]
}
```

Enumerations: `estimate_type` ∈ {observed, scaled}; `quality` ∈ {good, degraded, unknown, observed-zero}; `availability state` ∈ {available, degraded, missing, ended}; `flag` ∈ {unusually-high, unusually-low, typical, insufficient-history, not-comparable}; `season.status` ∈ {ok, unavailable}. Dates and numbers in the example are illustrative, not measured.

A.4 fireatlas-sensor-notices-v1 (fireatlas/samples/sensor_notices.json)

```
{
  "schema": "fireatlas-sensor-notices-v1", "method_version": "1",
  "notices": [
    {"id": "snpp-2024-07-processing-stop", "source_id": "VIIRS_SNPP_SP",
      "start_utc": "2024-07-24T05:28:00Z", "end_utc": "2024-07-28T00:00:00Z",
      "type": "outage", "url": "...", "retrieved_utc": "...", "quote": "..."}
  ]
}
```

A.5 fireatlas-replay-step-v1 (Tier 2, GET /api/replay/park-2024/step?as_of=...)

Fields: `as_of_utc`, `inputs_used[]` (id, kind, valid_at, available_at, age_minutes), `ensemble` (members, seed_base, horizons_h: [6,12,24], `burn_probability_tiles` URL template), `sectors[]` (id, rank, score, U, E, S, V, reason, geometry), `assimilation` (applied, perimeter_id, ess), `outcome` (present only when `reveal=true`: next_perimeter_id, jaccard, brier, hit_rate_top5), `limits[]`, `banner`.

Appendix B — Glossary (plain words)

Term	Meaning
Active-fire detection / hotspot	A satellite pixel flagged as containing heat from a fire. Not the fire's size or edge.
MODIS	Sensor on NASA's Terra (2000–) and Aqua (2002–) satellites; ~1 km fire pixels.
VIIRS	Sensor on Suomi NPP (2012–2026), NOAA-20 (2018–) and NOAA-21 (2022–); 375 m fire pixels, sees more small fires.
FIRMS	NASA's Fire Information for Resource Management System, which distributes both.
Standard vs NRT	Standard (science-quality, weeks-months later) vs near-real-time (hours). Only standard products build baselines.
Cell-day	One 1 km grid cell with ≥ 1 detection on one day. Makes pixel sizes more comparable.
Harmonized value	The activity a day would show in S-NPP VIIRS terms, estimated when S-NPP was not flying or missing.
$r(m)$	The ratio of VIIRS to MODIS cell-days in calendar month m during the years both flew.
LOYO	Leave-one-year-out: hide a year, fit on the rest, test on the hidden year.
Bootstrap interval	Range from recomputing the ratio on many resampled sets of years; shows uncertainty.
Degraded / missing day	A day when a sensor was partly or fully unavailable according to NASA notices or granule counts.
Critical period	≥ 5 days in a row when the 15-day activity was above the normal 90th percentile.
Burn probability	Share of simulated scenarios in which a place burned. Not a forecast.
Entropy (U)	How much the scenarios disagree about a place: 0 = all agree, 1 = split 50/50.
TFR	FAA Temporary Flight Restriction over an incident.

Appendix C — File map after Tier 1 (target)

Path	Role
fireatlas/regions.py	Preset regions, bboxes, fire regime, official links
fireatlas/archive.py	FIRMS archive import (multi-region)
fireatlas/aggregates.py	Daily cell-day aggregates (8 variants)
fireatlas/availability.py + granules.py	CMR overpass ledger + notices
fireatlas/calibration.py	r(m), q(m), bootstrap, LOYO, stability
fireatlas/calendar_v2.py	Calendar v2 API logic (harmonized series, climatology, seasons)
fireatlas/reference_mcd64.py	MCD64A1 burned-area comparison
fireatlas/validation_check.py	Independent recount of harmonized values
fireatlas/briefing.py	Season context card
fireatlas/sources.py	Single list of datasets for the Sources page and tests
fireatlas/web.py	HTTP server and routes
fireatlas/static/index.html, app.js, calendar-heatmap.js, calendar.css	Home calendar
fireatlas/static/method.html/js/css	Method & validation
fireatlas/static/data.html/js	Sources page
fireatlas/samples/aggregates/, calibration/, availability/, sensor_notices.json	Published data products
scripts/launch_demo.sh, smoke_test.sh, export_static.py	Run, verify, publish
docs/HARMONIZATION_METHOD.md, DATA.md, AI_USE.md, TEAM.md	Documentation
docs/winning-plan/SCORECARD.md	Live tracker

Tier 2 adds: fireatlas/replay.py, terrain.py, spread.py, sectors.py, observations_aerial.py, assimilate.py, fireatlas/static/replay.html/js/css.

Appendix D — SCORECARD snapshot at build time

The live version is `docs/winning-plan/SCORECARD.md`; rebuild this PDF to refresh the snapshot.

FireAtlas SCORECARD — live accountability tracker

Plan: `docs/winning-plan/FireAtlas_Winning_Plan.pdf` (source `docs/winning-plan/src/`). **Rules:** see plan section 9.1. Executors update **task rows** with evidence; only the **Auditor** (plan 9.2) changes **scores**. Every change adds a Changelog line. **Status words:** `NOT STARTED` · `IN PROGRESS` · `BLOCKED` (reason) · `DONE` (evidence: ...) · `DROPPED` (reason)

Field	Value
Last audit	2026-09-29 (baseline audit)
Audit score (100)	31.8 · Tier 1 target 54.2 · Tier 1 + 2 target 88.0
Local rubric (118)	≈65 · target ≈106
Branch	UNDECIDED (Gate 0 pending)
Tier 1 freeze date	2026-11-08 (Branch A)
Deadline risk	GREEN (no work scheduled yet slipped)
Public repo URL	—
Public demo URL	—

1. Gates

Gate	Status	Evidence
G0-T1 Written ruling from Local Lead	NOT STARTED	
G0-T2 Branch chosen (A/B)	NOT STARTED	
G0-T3 Official statement copied (28 Oct)	NOT STARTED	
Gate 1 (C12-T1...T3) repo integrity	NOT STARTED	
Tier 1 complete (all C1, C2, C10-T1...T7 DONE)	NOT STARTED	

2. Requirement trace (fill after G0-T3)

#	Requirement sentence (verbatim from the 2026 statement)	Feature / task that satisfies it	Status
1			

3. Category scores (Auditor only)

points = weight × score ÷ 5. Hard caps: synthetic-only ≤ 2 · not pushed ≤ 2 · no method + test ≤ 2 · forecast without held-out evaluation ≤ 3 · operational instruction ⇒ 0.

#	Category	Wt	Score (0-5)	Points	Target T1	Target T1+2	Evidence (file:line / command)
C1	Challenge fit and working calendar	12	3	7.2	5	5	Calendar works for one bbox, 2022–2026 authentic data; UTC only; no multi-year heatmap (<code>core.py:328-418</code>)
C2	Authentic NASA data, provenance, harmonization	12	3	7.2	5	5	30,823 FIRMS standard rows with hashes; no calibration; type filter missing (<code>core.py:310-325</code>); pipeline uncommitted
C3	Terrain, fuels, vegetation, dated weather	8	1	1.6	2	4	Visual-only elevation layer (<code>terrain-earth.html:32</code>)
C4	Explainable sector ranking	14	0	0.0	0	4	Absent
C5	UAS feasibility and airspace/command boundaries	7	1	1.4	2	4	Documentation mentions only
C6	Dated aerial observations and state update	8	0	0.0	0	4	Absent
C7	Spread scenarios, uncertainty, freshness	10	1	2.0	1	4	<code>research.py:266</code> states spread unavailable
C8	Containment boundaries and operational safety	7	2	2.8	3	4	Limits text exists; no banned-phrase test
C9	Responder workflow, resident info, alert separation	6	2	2.4	3	4	Briefing card with fixed label (<code>briefing.py:42-45</code>)
C10	Validation, replay, leakage, baselines	8	2	3.2	4	5	Recount verifier + CAL FIRE association 7/25; MODIS masks 0/17
C11	Interface clarity and demo reliability	4	3	2.4	5	5	Works locally; cluttered; ArcGIS globe failed to load

#	Category	Wt	Score (0-5)	Points	Target T1	Target T1+2	Evidence (file:line / command)
C12	Build, docs, reproducibility	4	2	1.6	5	5	81 tests pass; no LICENSE; repo 2 commits behind; ~40 untracked files
	Total	100		31.8	54.2	88.0	Arithmetic: $7.2+7.2+1.6+0+1.4+0+2.0+2.8+2.4+3.2+2.4+1.6 = 31.8$

4. Local rubric estimate (Auditor only)

#	Criterion	Max	Now	Target	Justification (one sentence)
1	Impact	20	10	17	Clear problem, one region only, no user evidence
2	Creativity	20	11	16	Evidence tracing is distinctive; calendar view conventional
3	Validity	20	11	18	Authentic data; no calibration or held-out test yet
4	Relevance	20	11	19	Harmonization not implemented; off-topic features present
5	Presentation	20	12	18	Polished visuals; no video/slides; story diluted
6	Teamwork	5	2	5	Two committers; no PR/review evidence
7	User experience	5	3	5	Usable but crowded
8	NASA data usage	5	4	5	FIRMS central; versions not surfaced in UI
9	Challenge category named	1	1	1	
10	Repository access	1	0	1	Latest work not pushed
11	Project page complete	1	0	1	Not submitted
	Total	118	≈65	≈106	

5. Task status

ID	Task	Status	Evidence (commit · test line · screenshot/JSON)	Updated
G0-T1	Written ruling	NOT STARTED		
G0-T2	Choose branch	NOT STARTED		
G0-T3	Re-read statement 28 Oct	NOT STARTED		
C12-T1	Commit + push, repo public	NOT STARTED		
C12-T2	Launcher, smoke test, CI	NOT STARTED		
C12-T3	Declare dependencies	NOT STARTED		
R1	Delete disaster casebook	NOT STARTED		
R2	Archive + remove training lab	NOT STARTED		
R3	Remove PWA	NOT STARTED		
R4	Remove synthetic demo UI	NOT STARTED		
R5	Replace synthetic tour	NOT STARTED		
R6	Remove EONET from home	NOT STARTED		
R7	Demote NRT globe	NOT STARTED		
R8	Remove ArcGIS globe from home	NOT STARTED		
R9	Delete earth.html + terrain HTML	NOT STARTED		
R10	Merge Research Lab into method	NOT STARTED		
R11	HMS out of calendar switch	NOT STARTED		
R12	Move NRT dropdown	NOT STARTED		
R13	Delete stale presentation docs	NOT STARTED		
R14	Move rubric JPGs	NOT STARTED		
R15	Move plan PDF	NOT STARTED		
R16	Replace hardcoded S-NPP gap	NOT STARTED		
R17	README/PRD rewrite trigger	NOT STARTED		
C2-T1	Multi-region importer	NOT STARTED		
C2-T2	Daily aggregates (8 variants)	NOT STARTED		
C2-T3	Availability ledger	NOT STARTED		
C2-T4	Filters + version rules	NOT STARTED		
C2-T5	Calibration + LOYO	NOT STARTED		
C2-T6	Provenance + citations	NOT STARTED		
C1-T1	Calendar API v2	NOT STARTED		
C1-T2	Multi-year heatmap	NOT STARTED		
C1-T3	Climatology panel	NOT STARTED		
C1-T4	Seasons + critical periods	NOT STARTED		

ID	Task	Status	Evidence (commit · test line · screenshot/JSON)	Updated
C1-T5	Area selection	NOT STARTED		
C1-T6	Day drawer	NOT STARTED		
C1-T7	Insight templates	NOT STARTED		
C1-T8	Exports	NOT STARTED		
C10-T1	Publish LOYO validation	NOT STARTED		
C10-T2	2012 step test	NOT STARTED		
C10-T3	MCD64A1 comparison	NOT STARTED		
C10-T4	HMS cross-check	NOT STARTED		
C10-T5	MODIS mask fix + review	NOT STARTED		
C10-T6	Recount verifier v2	NOT STARTED		
C10-T7	Leakage rules	NOT STARTED		
C11-T1	Home structure	NOT STARTED		
C11-T2	Remove visual noise	NOT STARTED		
C11-T3	Sensor timeline	NOT STARTED		
C11-T4	Method page sections	NOT STARTED		
C11-T5	4-step tour	NOT STARTED		
C11-T6	Empty/error states	NOT STARTED		
C11-T7	Reliability + Lighthouse	NOT STARTED		
C11-T8	5-person usability test	NOT STARTED		
C12-T4	Licence + image sources	NOT STARTED		
C12-T5	README + PRD	NOT STARTED		
C12-T6	Static site + Release asset	NOT STARTED		
C12-T7	Sources page + AI use	NOT STARTED		
C12-T8	Teamwork evidence	NOT STARTED		
C12-T9	Release tag + hash check	NOT STARTED		
C8-T1	Safety wording + banned-phrase test	NOT STARTED		
C9-T1	Season context + official links	NOT STARTED		
C5-T1	Aerial-observation scope text	NOT STARTED		
S-T0	Replay clock + leakage guard	NOT STARTED		
C3-T1	Terrain	NOT STARTED		
C3-T2	Fuels	NOT STARTED		
C3-T3	Dated weather	NOT STARTED		
C7-T1	Spread model	NOT STARTED		
C7-T2	50-member ensemble	NOT STARTED		
C7-T3	Freshness + uncertainty display	NOT STARTED		
C4-T1	Sectors	NOT STARTED		
C4-T2	Score components	NOT STARTED		
C4-T3	Explanation UI	NOT STARTED		
C4-T4	Retrospective hit rate	NOT STARTED		
C6-T1	Aerial perimeter ingest	NOT STARTED		
C6-T2	Assimilation	NOT STARTED		
C6-T3	Replay metrics	NOT STARTED		
C5-T2	Airspace boundaries panel	NOT STARTED		
C5-T3	Feasibility numbers	NOT STARTED		
C8-T2	Replay safety design	NOT STARTED		
C9-T2	After-action workflow	NOT STARTED		
C9-T3	Resident separation	NOT STARTED		
SUB-T1	Project page	NOT STARTED		
SUB-T2	30-second video	NOT STARTED		
SUB-T3	7 slides	NOT STARTED		
SUB-T4	Demo run sheet + rehearsals	NOT STARTED		

6. Data inventory (fill from C2-T1)

Region	Product	Years	Rows	Complete months	SHA-256
norcal	MODIS_SP	2022-07 → 2026-06	8,151	—	(from bundle manifest)
norcal	VIIRS_SNPP_SP	2022-07 → 2026-06	22,672	—	(from bundle manifest)

7. Ground truth and measured results

Measure	Value	Command	Date
Test suite	81 tests OK	<code>uv run python -m unittest discover -s tests</code>	2026-09-29
Park 2024-07-25 MODIS	603 pixels → 415 cells; S-NPP 0 rows	<code>/api/harmonization</code>	2026-09-29
Smoke test calendar value (C12-T2)	—	<code>scripts/smoke_test.sh</code>	
Calibration results (C2-T5): region · r_all · LOYO median abs log error (MoY / single / none) · coverage · beats baselines	—	<code>python -m fireatlas.calibration --region ...</code>	
Step test 2012 (C10-T2)	—		
MCD64A1 Spearman (harmonized / naive / MODIS-only)	—		
HMS cross-check Spearman	—		
MODIS mask reconciliation rate	0/17 processed	<code>python3 -m fireatlas.masks --case grove-2025</code>	2026-09-29
Lighthouse (Perf / A11y)	—		
Usability test (tasks passed / 5)	—		
Replay: Jaccard / Brier vs persistence / no-assimilation	—		
Replay: top-5 hit rate vs random range	—		

8. Changelog

When (UTC)	Who	What changed	Evidence
2026-09-29	Lead engineer (audit)	Baseline scorecard created from audit	Audit report, 81 tests OK